

GENERAL ORGANIC CHEMISTRY (GOC PART-2)

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THEORY

1. GENERAL ORGANIC CHEMISTRY

1.1 Introduction

In 1807, Berzelius proposed the term 'Organic Chemistry' for the study of compounds derived from natural sources. This was based on the theory of **vitalism** which said that all living systems possessed a 'vital force' which was absent in non-living systems. Compounds derived from living natural sources (organic) were thought to be fundamentally different from inorganic compounds.

Vital
Force

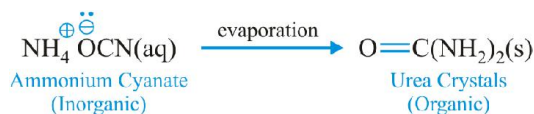
Living Systems

No Vital
Force

Non-living Systems

The vital force could be philosophically thought as the mysterious force God instilled in the living systems.

In 1823, Friedrich Wohler joined Berzelius as his student. In 1828, Wohler made a discovery which changed the definition of organic chemistry. Wohler conducted the following experiment.



Wohler successfully synthesized an organic compound starting from an inorganic compound. Following this, many others synthesized organic compounds starting from inorganic compounds. Thus, the theory of vitalism and the definition of organic chemistry lost its meaning.

But what was common in all the above compounds synthesized was the presence of carbon. Carbon shows a special property **catenation**. Carbon can connect with other carbon atoms to form long chains and rings (self-catenation) and can connect with atoms of many other elements in the periodic table (cross-catenation). Because of this reason, carbon can form a wide variety of compounds. Therefore, **the modern definition of organic chemistry is the study of carbon compounds.**

Probably, the vital force can be explained by the fact that most of the life-giving and life-sustaining functions are performed by carbon compounds, for example, the human tissues and skin are formed by proteins, respiration is possible due to haemoglobin, the information in our genes is carried out in the form of DNA/RNA etc.

General Organic Chemistry is the detailed study of the basic concepts and factors that govern the progress and outcome of reactions.

Note : The making and breaking of bonds usually occurs in several discrete steps before transforming transforming into products. The detailed sequential description of all the steps is called the **mechanism** of the reaction.

2.1 Hybridization

Hybridisation is a process in which two or more atomic orbitals of comparable energy of the valence-shell of an atom (**central atom** of the molecule or ion) either in its ground state or in its excited state mix together and give rise to the formation of new degenerate orbitals which are called **hybrid orbitals**.

2.2 Applications of Hybridization

Hybridization	% s-character
sp^3	25.0
sp^2	33.3
sp	50.0

2.2.1 Size of Hybrid Orbitals

As % s-character increases, size of hybrid orbital decreases. Therefore

$$\text{Size of Hybrid Orbital : } sp^3 > sp^2 > sp$$

2.2.2 Electronegativity of Hybrid Orbitals

As % s-character increases, electronegativity of hybrid orbital increases. Therefore

$$\text{EN of Hybrid Orbital : } sp > sp^2 > sp^3$$

2.3 Dienes

Dienes are organic compounds containing two double bonds. There are three types of dienes :

(a) Isolated (b) Conjugated (c) Cumulated

2.3.1 Isolated Diene

In this case, double bonds are separated by atleast one sp^3 carbon.

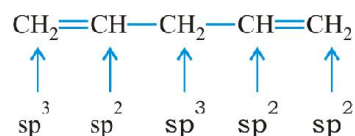
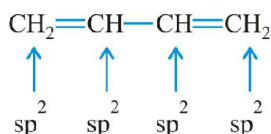


Table : Hybridization of Common Molecules.

Molecule	σ	π	LP	Hybridization	Geometry	Shape
*CH ₄	4	0	0	sp ³	Tetrahedral	Tetrahedral
*NH ₃	3	0	1	sp ³	Tetrahedral	Trigonal Pyramidal
H ₂ O*	2	0	2	sp ³	Tetrahedral	V-shape or Bent
*CH ₂ =CH ₂	3	1	0	sp ²	Trigonal Planar	Trigonal Planar
*CH≡CH	2	2	0	sp	Linear	Linear

2.3.2 Conjugated Diene

Double bonds are separated by only one single bond (or 4 sp² carbons in a row).



2.3.3 Cumulated Diene

Both sets of double bonds are at the same carbon.



A substituted **allene**

An allene is CH₂=C=CH₂

2.3.4 Stability of Dienes

The relative stabilities of dienes follows the order

Conjugated > Isolated > Cumulated

Important :

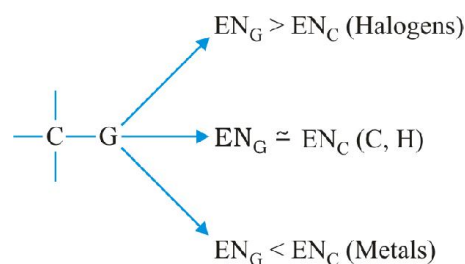
$$\text{Stability} \propto \frac{1}{\text{Energy Content of the molecule}}$$

Note : Diamond - Each C is sp³. Tetrahedral solid.

Graphite - Each C is sp². Layered solid with weak van der Waal's forces between layers.

Calcium Carbide - Each C is sp.

Fullerene - Each C is sp².



There are broadly three types of groups/atoms that may be attached to carbon as illustrated. Although C is more electronegative than H, the electronegativity difference is small and the bond is generally considered non-polar.

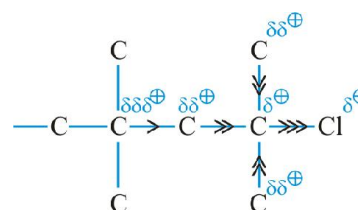
3.1 Nature of Inductive Effect

Inductive effect is a **permanent effect** and can be directly correlated to its dipole moment.

It is a weak effect as the shifting of electrons takes place only through sigma bonds.

3.2 Effect of branched carbon chain

An illustration has been marked for operation of inductive effect which is self-explanatory.



More the number of δ , lesser the effect

3. INDUCTIVE EFFECT

When two unlike atoms form covalent bond, the electron-pair forming the sigma bond is never shared equally between the two atoms but is shifted slightly towards the more electronegative species.

3.3 Electron Donating and Electron withdrawing Groups

Inductive effect may be due to single atom or a group of atoms. Relative inductive effects are measured with reference to hydrogen. Those which donate electrons to

carbon chain are called **electron-donating groups (EDG)** or **electron-releasing groups (ERG)** and are said to exert **+I effect**. Those which withdraw electrons from carbon chain are called **electron-withdrawing groups (EWG)** and are said to exert **−I effect**.

Important :

1. I.E. of alkyl groups : $3^\circ > 2^\circ > 1^\circ > \text{CH}_3-$
2. In general, greater is the number of carbons in an alkyl group, greater is its +I effect.
3. For problem-solving, we take electronegativity of sp^3 hybridized carbon to be more than sp^3 hybridized nitrogen.

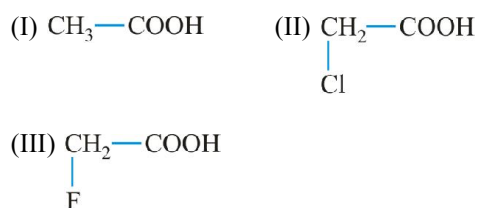
3.4 Applications of Inductive Effect

3.4.1 Effect on Acidic/Basic Strength

EWG increases acidic strength and decreases basic strength. ERG decreases acidic strength and increases basic strength.

Example - 1

Compare the acidic strength :



Solution :

An alkyl group is donating only if no other EWG is present on it. Therefore, groups like $-\text{CH}_2\text{Cl}$ and $-\text{CH}_2\text{F}$ become electron withdrawing groups.

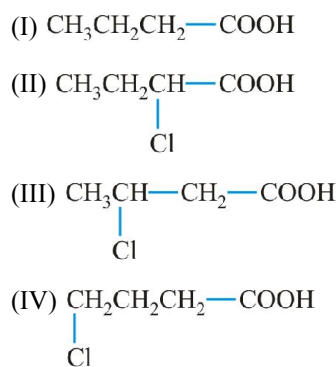
Order of Acidic Strength : $\text{III} > \text{II} > \text{I}$

3.4.2 Effect of Distance

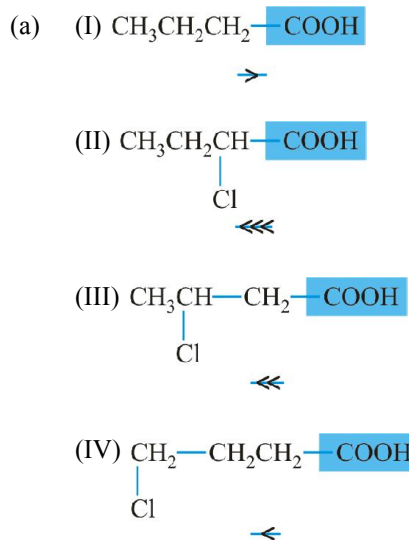
If the ERG/EWG moves away, the inductive effect diminishes.

Example - 2

- (a) Compare the acidic strength of :



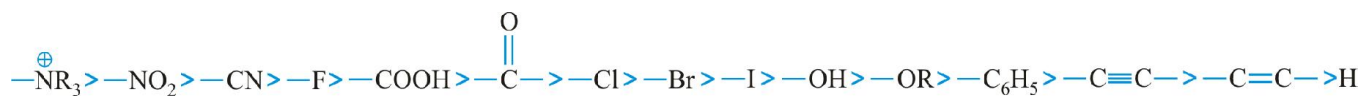
Solution :



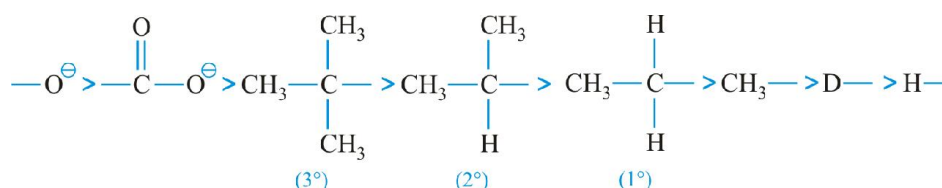
Order of acidic strength : $\text{II} > \text{III} > \text{IV} > \text{I}$

Series of +I and −I groups in order of their strength

−I Series (EWG)

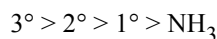


+I Series (ERG)

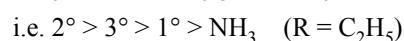
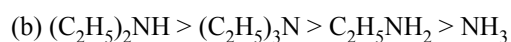
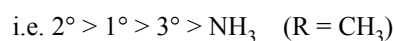
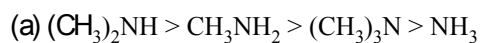


3.4.3 Basicity of Amines

The order in **gaseous or liquid phase** is :



To determine the basic strength of amines in **aqueous phase**. We have to consider **inductive effect, solvation effect and steric hinderance**. The order of basic strength is therefore experimental in aqueous state as we can't give priority to stability provided by any one factor. Two results are important for aqueous phase :



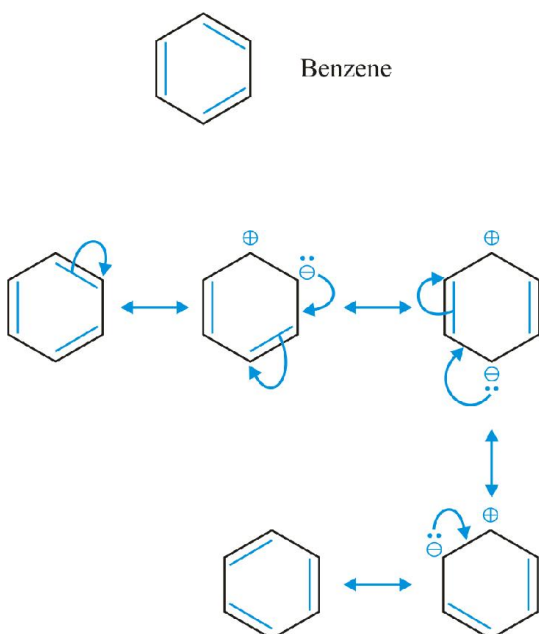
4. RESONANCE

Molecules are generally represented by simple lewis structures but some molecules can not be represented by just one Lewis structure. This led to the discovery of resonance. Resonance refers to the delocalization of electrons (generally π -electrons).

4.1 Conjugated Systems

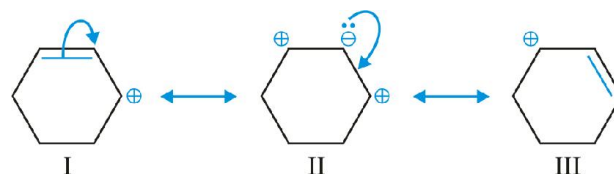
4.1.1 Pi alternate Pi

Example - 3



4.1.2 Pi alternate Positive Charge

Example - 4



4.1.3 Pi alternate Negative Charge

Example - 5



4.1.4 Pi alternate Odd Electron

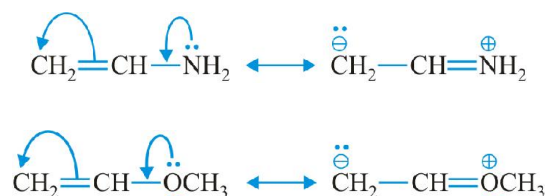
Example - 6



4.1.5 Pi alternate Lone Pair

This case is similar to 'pi alternate negative charge' as lone pair and negative charge are treated similarly.

Example - 7



4.1.6 Lone Pair and Positive Charge on Adjacent Atoms

Example - 8



4.2 Rules for Validity of Lewis Structures

Rule-1 :

All the lewis structures must conform to lewis octet rule.

Rule-2 :

Position of atoms in all resonating structures must be the same. Only the electrons move.

Rule-3 :

All the resonating structures must have the same number of paired and unpaired electrons, i.e. sum of bond pairs and lone pairs must be constant.

Rule-4 :

All the atoms participating in resonance in a molecule must be coplanar. This is required for the effective overlap of p orbitals and the delocalization of electrons, for eg, buta-1,3-diene.

4.3 Criteria for Major/Minor Contributors

Resonance forms can be compared using the following criteria in the following order :

1. As many octets as possible (a neutral molecule is always more stable which its octet is complete).
2. As many π bonds as possible.
3. Negative charge on more electronegative atom is stable.
4. Charge separation.
 - (a) Similar charges - Keep them as FAR as possible to minimize repulsion and instability.
 - (b) Opposite charges - Keep them as NEAR as possible to maximize attraction and stability.

Example - 9

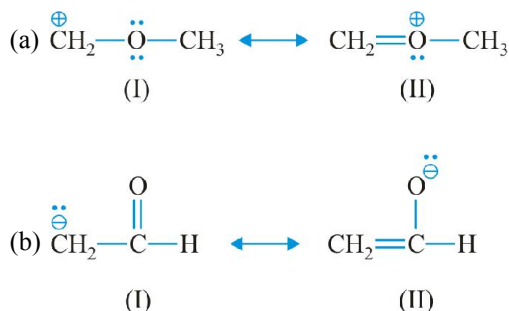
Which of the following structures is more stable ?

**Solution :**

II is more stable as all the octets are complete.

Example - 10

Which of the following is more stable in the following pairs ?

**Solution :**

- (a) In II, all octets are complete. Therefore, II is more stable.

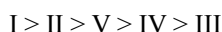
- (b) I and II are tied on octets and number of π bonds but negative charge is more stable on more electronegative atom. Hence, II is more stable.

Example - 11

Give the order of stability of following resonating structures

**Solution :**

In (I), there are maximum number of pi bonds. Therefore, it is most stable. In (II) and (V), the number of pi bonds is equal but charge separation is greater in (V). Therefore, (II) is more stable than (V). In (III) and (IV), there is maximum charge separation but (III) is highly unstable due to electrostatic repulsion. Hence, the order of stability is :

**5. MESOMERIC EFFECT**

The permanent polarization, due to a group conjugated with a π bond or a set of alternate π bonds, is transmitted through the π electrons of the system-resulting in a different distribution of electrons in the unsaturated chain.

This kind of electron redistribution in unsaturated compounds conjugated with electron-releasing or electron-withdrawing groups (or atoms) is called **Mesomeric Effect or Resonance Effect**.

This effect is **permanent** and is indicated by the dipole moment.

5.1 Electron-Releasing and Electron-Withdrawing Groups

Groups which release or withdraw electrons by resonance are said to exert M or R effect.

5.1.1 Electron-Releasing Groups (+R or +M effect)

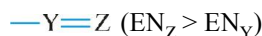
The common thing about all the groups listed is that the atom connected with the conjugated system has a lone pair to donate. Therefore, a generic representation can be



5.1.2 Electron-Withdrawing Groups (–R or –M effect)



The common thing about all the groups listed is that the atom connected with the conjugated system has a π bond with another more electronegative atom which withdraws the electrons or directly has a positive charge on them. Therefore, a generic representation can be



5.1.3 Dual Behaviour

Groups such as $-\text{N}=\text{O}$ are both electron-releasing and electron-withdrawing as illustrated.

Example - 12

As electron releasing group



As electron withdrawing group



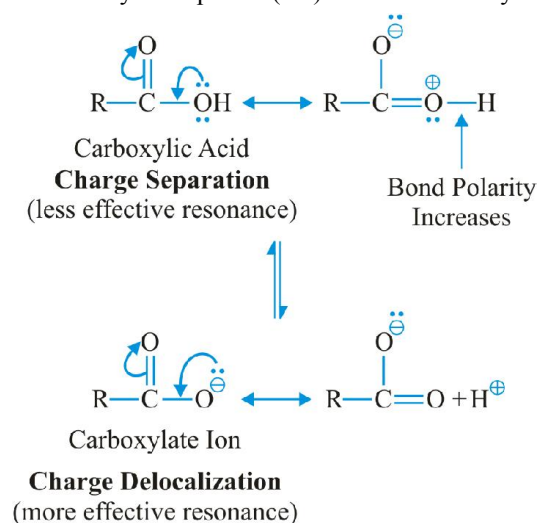
Which behaviour dominates and which is used in a particular context will be discussed later in **Electrophilic Aromatic Substitution** later.

Resonance Effect does NOT depend upon distance unlike inductive effect.

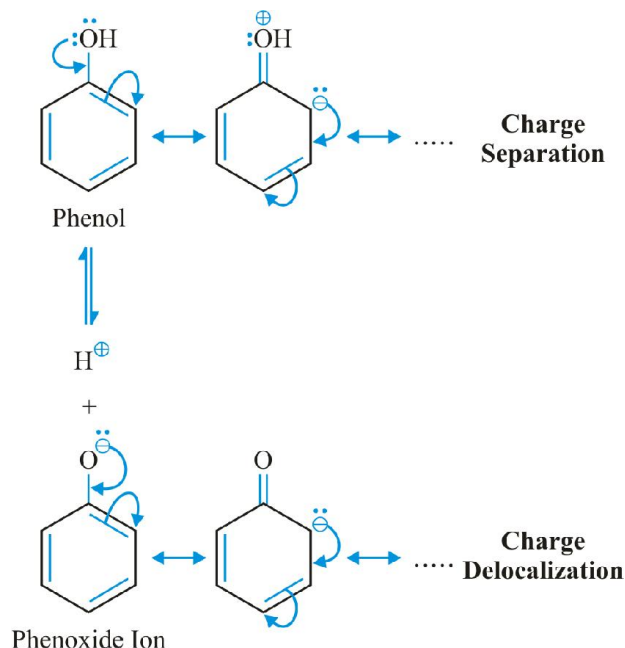
5.2 Applications of Mesomeric Effect

5.2.1 Effect on Acidic Strength of Carboxylic Acids and Phenols

The resonating structure of carboxylic acid leads to charge-separated structure which is less stable than the carboxylate ion in which charge is delocalized. Therefore, carboxylic acid readily loses proton (H^+) to form a carboxylate ion.



Similarly, in phenol, resonance leads to charge separation which increases the rate of ionization and forms phenoxide ion which is stabilized by charge delocalization.

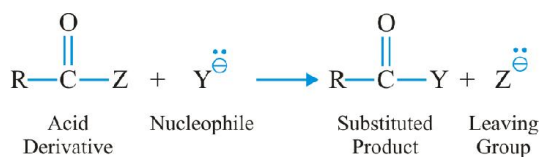


Note : Order of acidic strength



5.2.2 Effect on Reactivity of Carboxylic Acid Derivatives

A typical nucleophilic reaction is represented as :



The stronger is the bond between C and Z, the difficult it is for a nucleophile to break a bond and therefore, lower reactivity.



Reactivity order of carboxylic acid derivatives towards nucleophilic acyl substitution is :



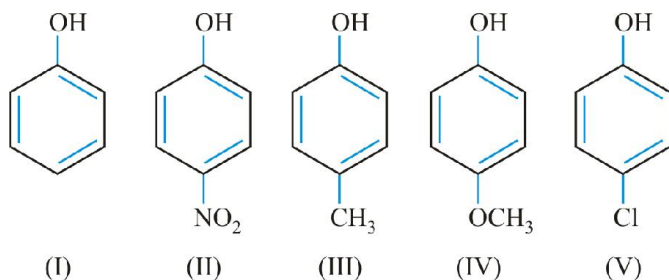
5.2.3 Effect of ERG/EWG on Acidic/Basic Strength

EWG increases the acidic strength and decreases the basic strength.

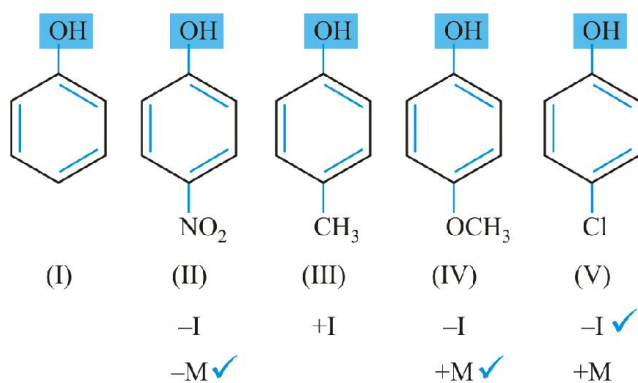
ERG decreases the acidic strength and increases the acidic strength.

Example - 13

Arrange the following in the order of decreasing acidic strength :

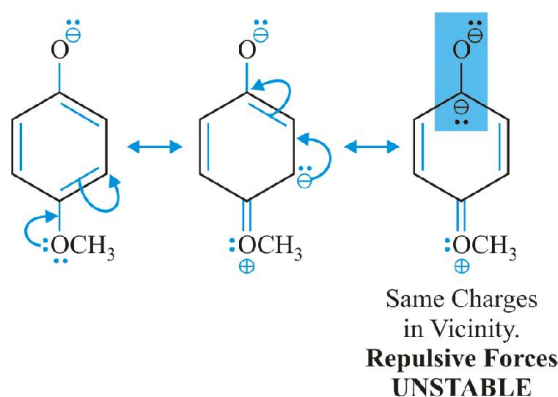
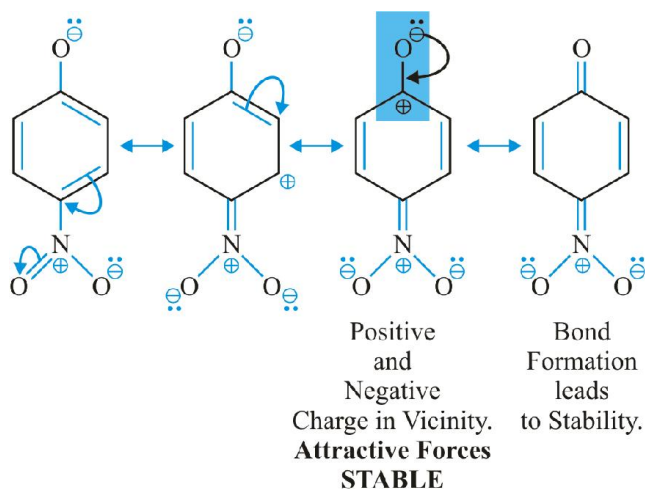


Solution :



The order of acidic strength is : II > V > I > III > IV

In the previous example, let's also discuss the stability of phenoxide ions corresponding to (II) and (IV).

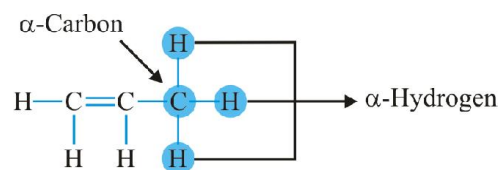


6. HYPERCONJUGATION

Hyperconjugation is the ability of the σ bond electrons of an α C—H bond to undergo conjugation with the adjacent π electrons. It is also known as **Baker-Nathan Effect**, **No-Bond Resonance** and σ - π Effect.

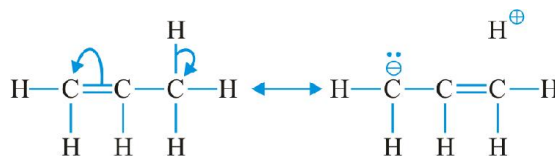
6.1 α -Carbon and α -Hydrogen

We have already discussed the α , β , γ nomenclature. Let's take an example :

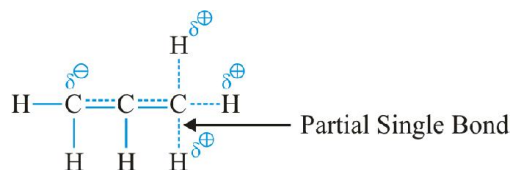


α -Carbon is the carbon attached to a functional group such as $C=C$. The hydrogen attached to α -carbon is called α -hydrogen. For an α C—H bond to be eligible for hyperconjugation, α C must be sp^3 hybridized.

6.2 Mechanism of Electron Donation in Hyperconjugation

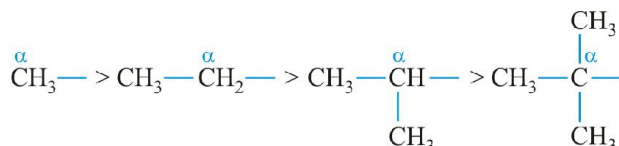


The hybrid formed by these resonating structures better known as hyperconjugating structures is :



Now, greater the number of α -H, greater the number of hyperconjugating structures and more is the electron donation of alkyl group to π bond.

The order of electron-donation of alkyl groups based on hyperconjugation is :



Methyl > 1° > 2° > 3°

Note : More is the number of α -H, more is the π bond delocalized. This implies that more will be the stability of alkene and less will be the heat of hydrogenation and more is the **no-bond resonance energy**.

6.3 Applications of Hyperconjugation

6.3.1 Stability of Alkenes

More is the number of α -hydrogen, more is the number of hyperconjugating structure and therefore more stability and greater no bond resonance.

Example - 14

Which alkene is more stable ?

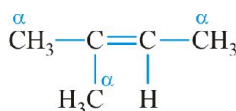


(I)



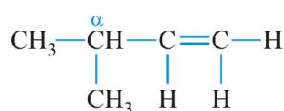
(II)

Solution :



(I)

$\alpha\text{C} = 3, \alpha\text{H} = 9$



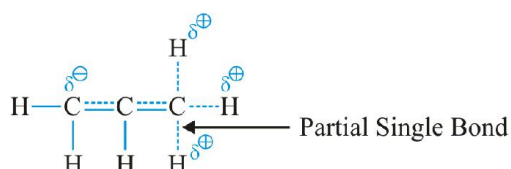
(II)

$\alpha\text{C} = 1, \alpha\text{H} = 1$

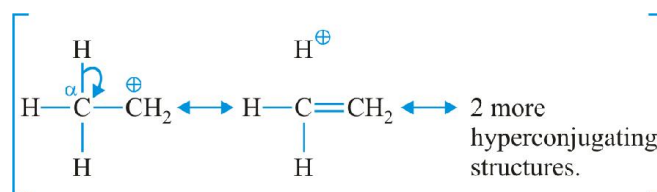
I is more stable than II.

6.3.2 Acidic Character of Alkenes

Hyperconjugation weakens the $\alpha\text{C-H}$ bond in hyperconjugation hybrid (partial single bond) and therefore αH can be lost easily.

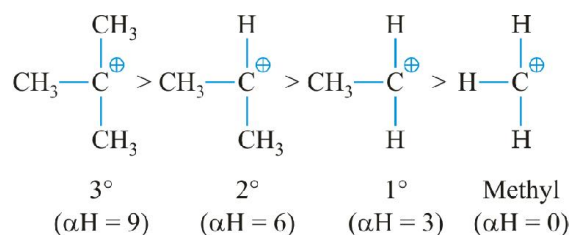


6.3.3 Stability of Carbocations



The positive charge on C is delocalized over αH to give stability to the carbocation. More is the number of αH , more is the stability of carbocations.

Stability of Carbocations



7. ELECTROMERIC EFFECT

Electromeric effect is observed only in the presence of a reagent and is therefore, a temporary effect. When a reagent approaches a molecule, the multiple bond such as $\text{C}=\text{C}$ or $\text{C}=\text{O}$ is polarized by the complete transfer of π electrons.



When the multiple bond is between two unlike atoms, the shift of electrons takes place towards more electronegative atom.



8. COMPARISON OF INDUCTIVE, HYPERCONJUGATION AND RESONANCE EFFECTS

Inductive Effect is a σ - σ interaction and acts through **strong sigma bonds**.

Resonance/Mesomeric Effect is a π - π interaction and acts through **weak pi bonds**.

Hyperconjugation is a σ - π interaction and acts through a **strong sigma and a weak pi bond**.

Therefore, the order of importance is :

Resonance > Hyperconjugation > Inductive

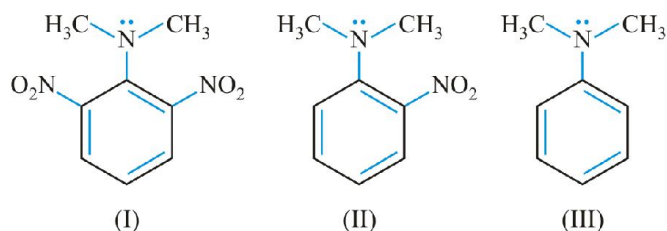
9. STERIC INHIBITION OF RESONANCE (SIR)

When both the **ortho** positions of a **bulky functional group** are occupied by **bulky substituents**, all the three groups are out of plane of the benzene ring.



Example - 15

Mark the order of basic strength :



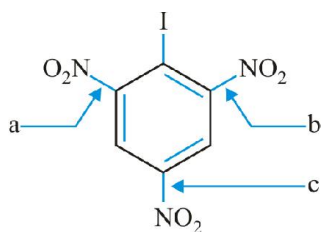
Solution :

In (II) and (III), the lone pair of N is in conjugation with the benzene ring and is not available for donation. (II) is less basic than (III) due to $-I$ and $-M$ of $-\text{NO}_2$ group. It may seem that (I) is least basic due to presence of 2 $-\text{NO}_2$ groups but $-\text{NO}_2$ and $-\text{N}(\text{CH}_3)_2$ are all bulky groups. This is a case of steric inhibition of resonance due to which the lone pair of N is not in conjugation and is readily available for electron donation. Hence, the order of basic strength is :

(I) > (III) > (II)

Example - 16

Mark the order of bond lengths in the given molecule.



Solution :

$-\text{I}$, $-\text{NO}_2$ are bulky groups and is case of steric inhibition of resonance. Therefore, the $-\text{NO}_2$ groups ortho to $-\text{I}$ are out of conjugation while the $-\text{NO}_2$ group para to $-\text{I}$ will be in conjugation with the benzene ring. Therefore, bonds 'a' and 'b' will always have single bond character while 'c' has double bond character. Therefore :

$c < a = b$

10. CARBOCATION

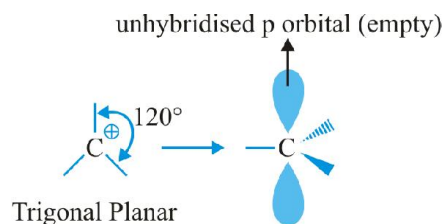
10.1 Definition

Carbocation is the intermediate of carbon containing positive charge. It has six electrons in the valence shell.

10.2 Geometry and Hybridization

Hybridization of $\text{C}^{\oplus} = \text{sp}^2$

Geometry of $\text{C}^{\oplus} = \text{Trigonal Planar}$



10.3 Classification of Carbocations

This classification will also be used for carbanions and carbon free radical and will be studied only in this section.

$\oplus\text{CH}_3$	Methyl Carbocation
$\text{R}-\oplus\text{CH}_2$	1° Primary Carbocation
$\text{R}_2-\oplus\text{CH}$	2° Secondary Carbocation
$\text{R}_3\text{C}^{\oplus}$	3° Tertiary Carbocation

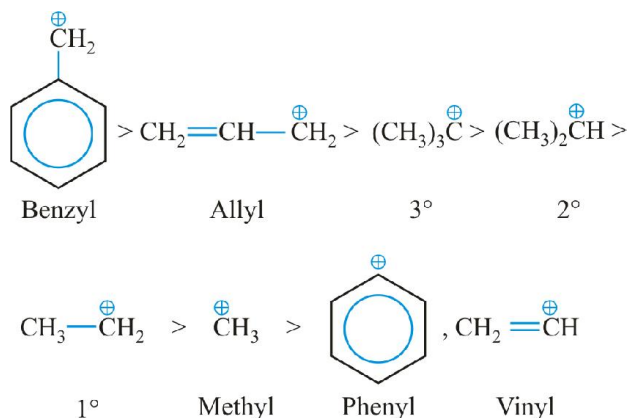
Type	Generic	Example
Allylic Carbocation		$\text{CH}_2=\text{CH}-\oplus\text{CH}_2$ Allyl Carbocation
Benzylic Carbocation		 Benzyl Carbocation
Vinylic Carbocation		$\text{CH}_2=\oplus\text{CH}$ Vinyl Carbocation
Aryl Carbocation		 Phenyl Carbocation

10.4 Stability of Carbocations

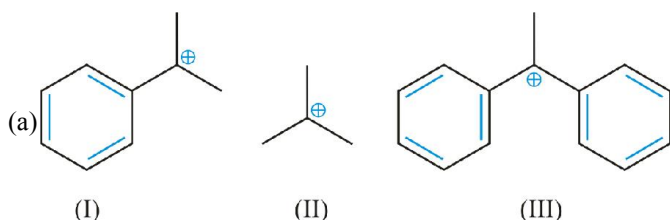
There are three factors contributing to the stability of carbocations :

- Inductive Effect
- Hyperconjugation
- Resonance

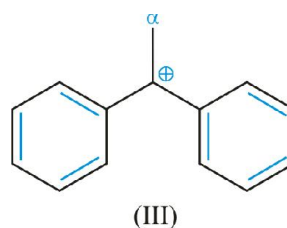
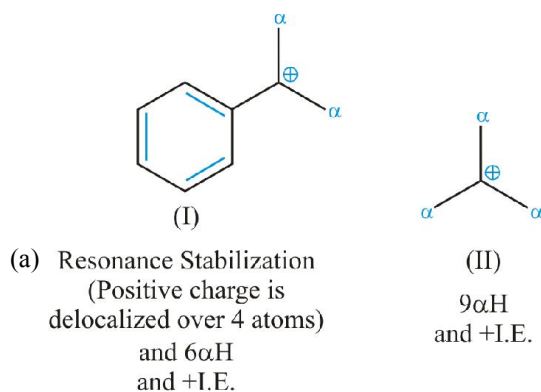
Order of stability :

**Example - 17**

Rank the stability of carbocations in each case :



Solution :



Resonance Stabilization
(Positive charge is
delocalized over 7 atoms
and $3\alpha\text{H}$
and +I.E.)

Order of stability : III > I > II

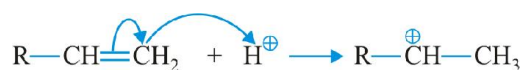
10.5 Formation of Carbocations**10.5.1 Ionization of Carbon-Leaving Group Bond**

In this method :

- Bond between carbon and leaving group ionizes.
- Leaving group accepts the pair of electrons that were shared in the covalent bond.

Rate of formation of carbocation depends on :

- The stability of carbocation formed.
- The nature of the leaving group. **Weaker the base better the leaving group.** This is because weaker leaving group implies a stable compound and its formation will therefore be favoured.

10.5.2 Addition of Proton to a π bond

Rate of carbocation formation depends on :

- Stability of carbocation formed.
- Strength of the electrophile.

10.6 Reactions of Carbocations

There are three important reactions of carbocations :

- Capture a Nucleophile
- Lose a proton to form a π bond.
- Rearrangement

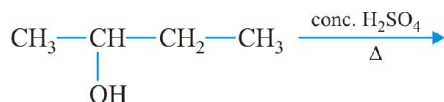
10.6.1 Capture a Nucleophile



10.6.2 Loose a proton to form a π bond

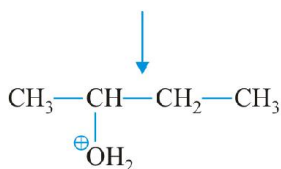
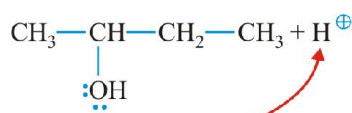
Example - 18

Form the products from the following reaction :

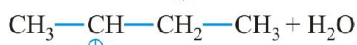
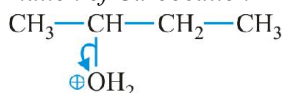


Solution :

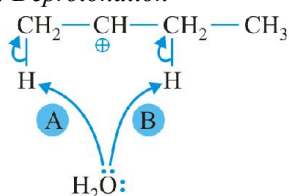
Step-1 : Protonation



Step-2 : Formation of Carbocation



Step-3 : Deprotonation



A
(2 α H)
Minor

B
(6 α H)
Major

When carbocation deprotonation can lead to more than one product, all products are formed and the most stable product is the major product.

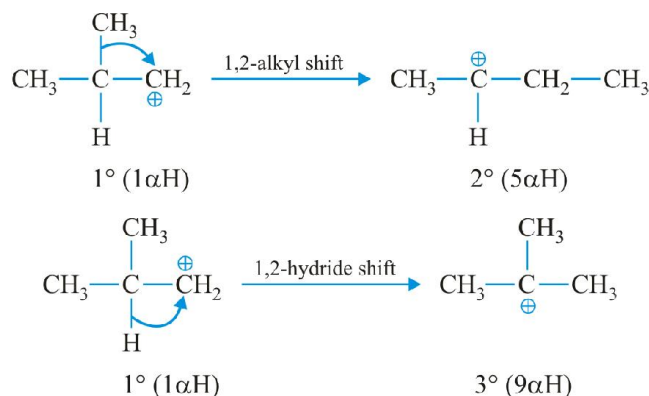
10.6.3 Carbocation Rearrangement

A carbocation can become more stable by rearrangement. Bonding electrons of carbocation may shift between adjacent atoms to form more stable carbocation. There are two kinds of shifts that take place in order to gain stability.

(a) Hydride Shift

(b) Alkyl Shift

Example - 19



In the above example, both hydride and methyl shifts are possible leading to more stable carbocation but **only that shift is preferred which leads to more stable carbocation**. In this example, hydride shift will take place.

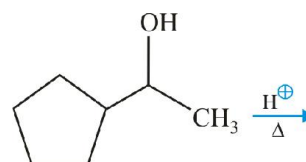
Important : The shift takes place in the form of $H^{\ominus}$ or $CH_3^{\ominus}$.

10.6.4 Cyclic Rearrangement/Ring Expansion

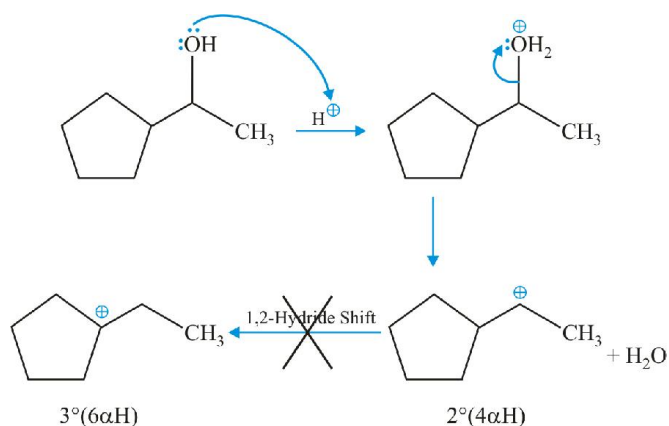
Ring expansion will take place only when positive charge is on carbon adjacent to the ring.

Example - 20

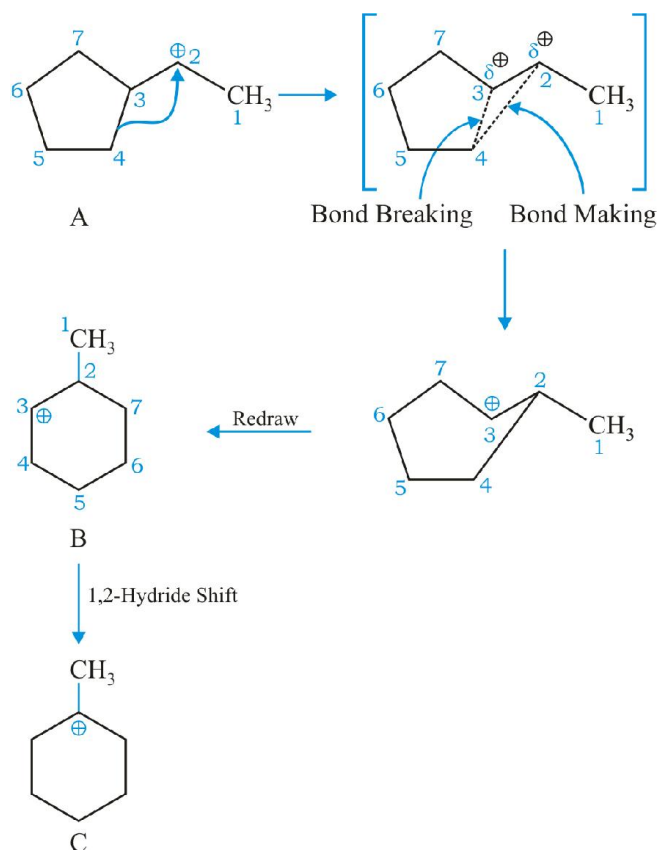
Draw the products of the following reaction :



Solution :



Even though the hydride shift leads to more stable carbocation, ring expansion is preferred.



11. CARBANIONS

11.1 Definition

Carbanion is the intermediate of carbon containing negative charge. It has eight electrons in the valence shell.

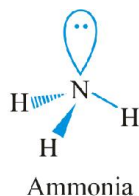
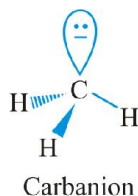


11.2 Geometry and Hybridization

Hybridization of C^- : sp^3

Geometry : Trigonal Pyramidal

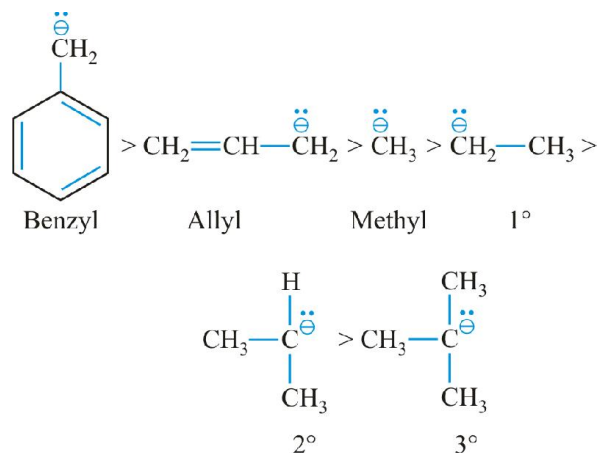
Carbanion and ammonia are isoelectronic species having same structure



11.3 Stability

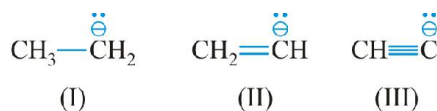
ERG will increase the electron density at carbon and will make it unstable. EWG will decrease the electro density at carbon and will make it stable.

Order of Stability :



Example - 21

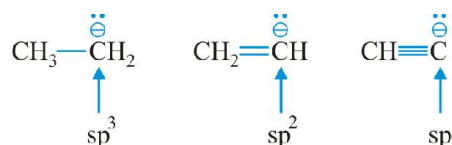
Give the order of stability of :



Solution :

As s-character increases, electronegativity of C increases and therefore negative charge will become more stable. Therefore, order of stability is :

$\text{III} > \text{II} > \text{I}$



11.4 Formation of Carbanion

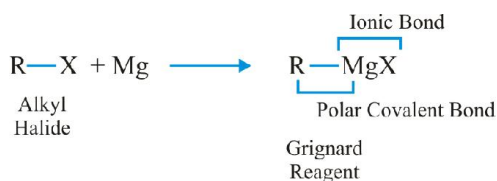
There are two methods for the formation of carbanion :

(A) Partial formation via Grignard Reagent

(B) Formation from Carbonyl Compounds

We will discuss these methods in the subsequent section

11.4.1 Formation via Grignard Reagent



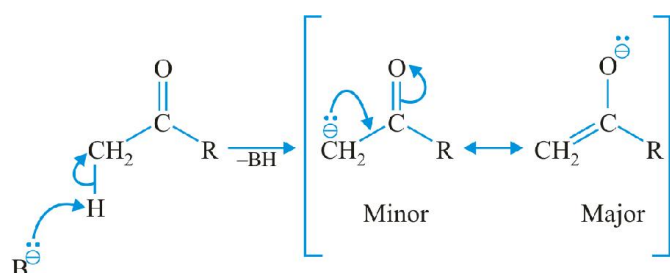
Note : $\text{R}-\text{MgX}$ Development of partial charges in polar covalent bond.

1. Carbanion is never formed as an intermediate that can be isolated in the case of Grignard Reagent. It directly participates in the reaction.
2. Dry ether is used in this formation as it is inert to Grignard reagent. For the formation of Grignard reagent from aryl halides, we use tetrahydrofuran (THF) as solvent.



Tetrahydrofuran (THF)

11.4.2 Formation from Carbonyl Compounds



There are three reasons for the easy formation of carbanion from carbonyl compounds

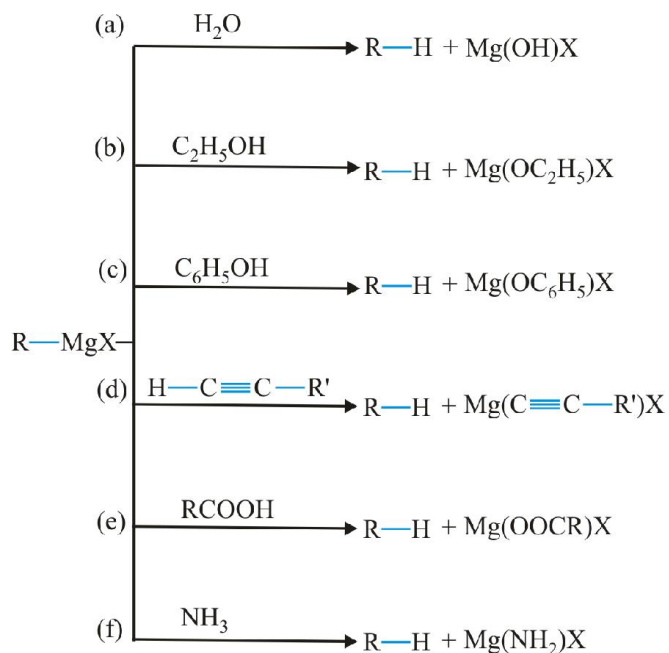
- (a) Resonance - stabilization of carbanion which is the conjugate base of carbonyl compound.
- (b) Hyperconjugation makes the $\alpha\text{C-H}$ bond acidic.
- (c) $-I$ of increases the acidic strength of C-H bond.

11.5 Reactions of Carbanion

The reactions of carbanion are very fast as electropositive carbon carries negative charge.

11.5.1 Grignard Reagent as a Base

In this reaction it captures acidic hydrogen.

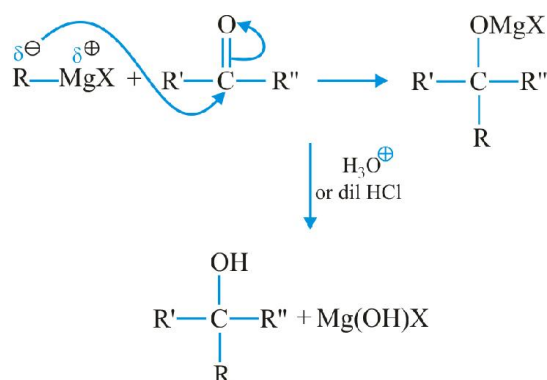


In (d), the reaction with terminal alkyne also takes place as sp -hybridized carbon is highly electronegative and therefore H attached to it is fairly acidic.

11.5.2 Grignard Reagent as a Nucleophile

Grignard Reagent reacts with carbonyl compounds to form alcohols. This is a very important method for the synthesis of alcohols.

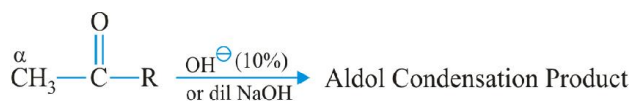
Mechanism :



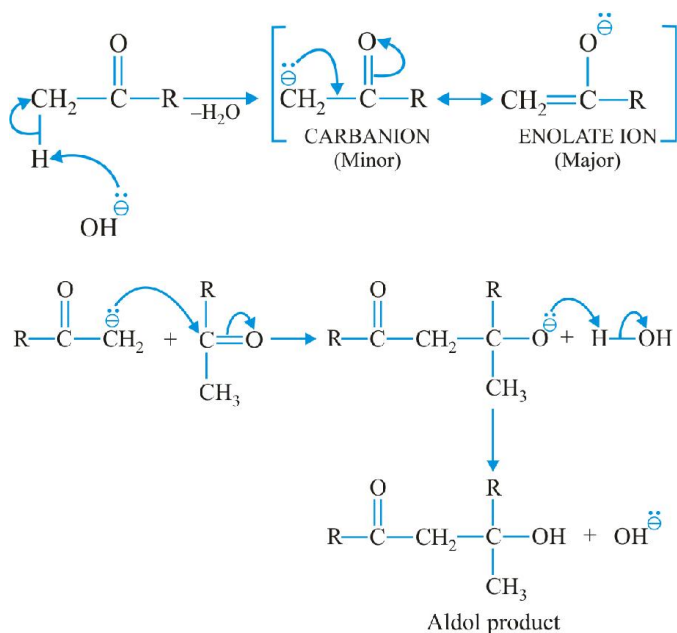
H_3O^+ is never added along with the first step as Grignard Reagent will react with H^+ to give $\text{R}-\text{H}$ as in previous reaction.

11.5.3 Aldol Condensation

This reaction is shown by carbonyl compounds containing atleast one α H in presence of dilute base such as dilute NaOH.



Mechanism :

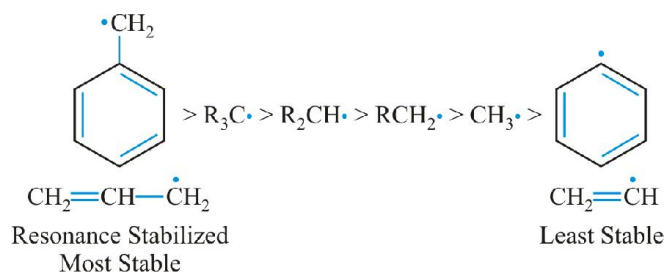


The hybridization of carbon free radical was proposed after experimental verification of structure of different radicals. It was proposed that when ERG are placed on $\text{C}^\bullet$, it has sp^2 hybridization and when EWG are placed on $\text{C}^\bullet$, it has sp^3 hybridization.

Molecule	Hybridization	Location of odd electron
$\bullet\text{CH}_3$	sp^2	p
$\bullet\text{CF}_3$	sp^3	sp^3

12.3 Stability

ERG increase stability while EWG decrease stability.



12.4 Formation of Carbon Free Radical

Carbon Free Radicals are formed by homolytic cleavage of bonds. They are formed :

- at high temperature in the gas phase
- in non-polar solvents
- by ultraviolet light
- by addition of other radicals

12.5 Reactions of Carbon Free Radical

The most common reactions in which free radical is involved are :

- Halogenation of alkanes.
- Addition of HBr in the presence of peroxides to alkenes. (Anti-Markonikov Rule)
- α -Halogenation of alkenes.
- Wurtz Reaction
- Decarboxylation reaction

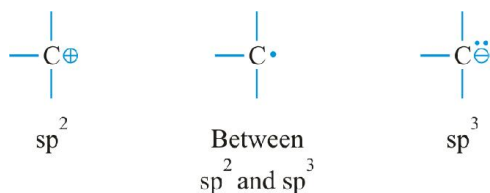
12. CARBON FREE RADICALS

12.1 Definition

Carbon Free Radical is the intermediate of carbon having an odd electron. It is neutral and has seven electrons in the valence shell. It is highly reactive as it requires only one electron to complete its octet and therefore, is short-lived.

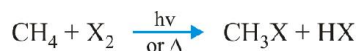


12.2 Geometry and Hybridization

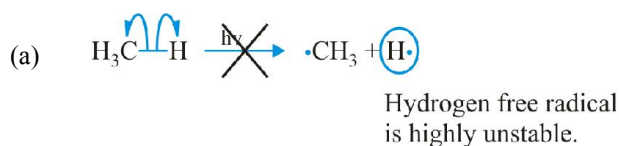


12.5.1 Free Radical Halogenation of Alkanes

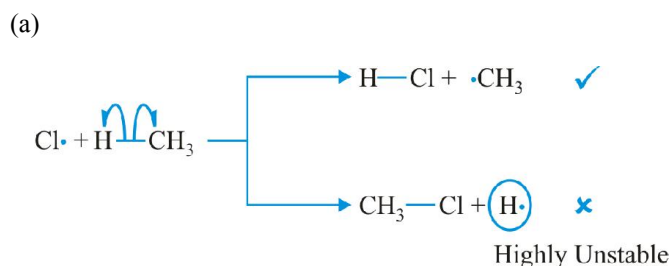
The typical reactions of free radical are chain reaction mechanisms. There are three steps in a chain reaction mechanism : initiation, propagation and termination.

**Mechanism****(I) Chain Initiation**

There are two choices :



(I) (b) will take place as it is energetically feasible.

(II) Chain Propagation

This will again participate in II(a) to propagate the reaction.

(III) Chain Termination

The various free radicals in the progress of reaction are : $\text{Cl}\cdot$, $\text{CH}_3\cdot$. There are ways in which free radicals from different chains may combine to terminate the reaction.



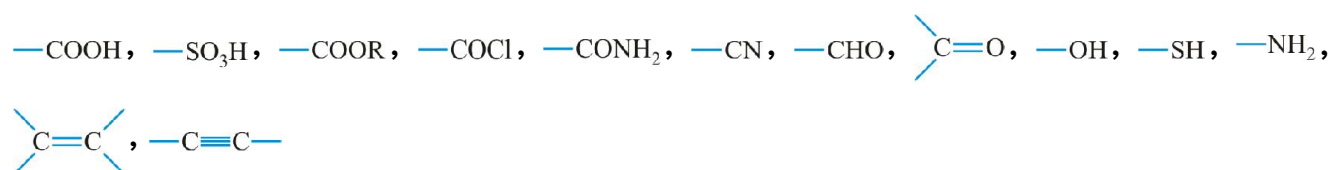
There is a side-product CH_3-CH_3 that is formed in this reaction. Besides these, a number of other side-products such as CH_2Cl_2 , CHCl_3 and CCl_4 have also been observed. These get formed since carbon free radical is highly reactive. It randomly reacts with other species that make it stabler.

13. FUNCTIONAL GROUPS

Class	Formula	Prefix	Suffix
Acid Halide	$-\text{COX}$	Halocarbonyl	- carbonyl halide
	$-(\text{C})\text{OX}$	-	- oyl halide
Alcohols	$-\text{OH}$	Hydroxy	- ol
Aldehydes	$-\text{CHO}$	Formyl	- carbaldehyde
	$-(\text{C})\text{HO}$	Oxo	- al
Amides	$-\text{CONH}_2$	Carbamoyl	- carboxamide
	$-(\text{C})\text{ONH}_2$	-	- amide
Amines	$-\text{NH}_2$	Amino	- amine

Class	Formula	Prefix	Suffix
Carboxylates	$\text{—COO}^\ominus$	Carboxylato	– carboxylate
	$\text{—(C)OO}^\ominus$	–	– oate
Carboxylic Acid	—COOH	Carboxy–	– carboxylic acid
	—(C)OOH	–	– oic acid
Ethers	—OR	Alkoxy–	–
Esters of Carboxylic Acid	—COOR	Alkoxy carbonyl	Alkyl .. carboxylate
	—(C)OOR	–	Alkyl ... oate
Ketones	>C=O	Oxo–	– one
Cyanides (Nitriles)	$\text{—C}\equiv\text{N}$	Cyano	– carbonitrile
	$\text{—(C)}\equiv\text{N}$	–	– nitrile
Salts of Carboxylic Acids	$\text{—COO}^\ominus \text{M}^\oplus$	–	Cation ... carboxylate
	$\text{—(C)OO}^\ominus \text{M}^\oplus$	–	Cation ... oate
Sulphonic Acids	$\text{—SO}_2\text{—OH}$	Sulfo–	– sulfonic acid
Thiols	—SH	Sulfanyl	– thiol

The order of decreasing priority for functional groups is :



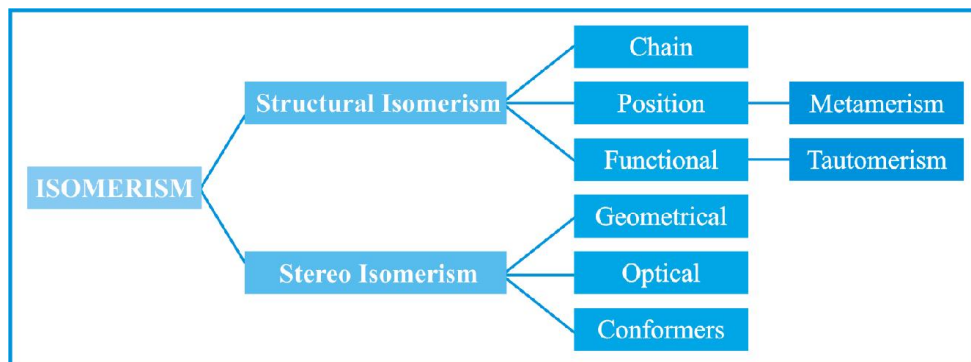
Important :

- The —R , $\text{—C}_6\text{H}_5$, halogens, nitro and alkoxy groups are always treated as substituents.
- If more than one functional group of same type is present, its number is indicated by adding di-, tri- etc.
- When C is marked in parenthesis, the carbon of the functional group is counted as part of the parent chain and the numbering will always begin from this carbon and its position need not be marked in the name.

14. ISOMERISM

In organic chemistry, a number of organic compounds having different physical and chemical properties can be represented by the same molecular formula. The property by virtue of which organic compounds possessing different physical and chemical properties and having the same molecular formula is known as isomerism and the different compounds are known as isomers.

Since the isomers are constituted of the same number of atoms, so it is clear that the difference in their properties must be due to the difference in the relative arrangement of the atoms within the molecules.

**15. STRUCTURAL ISOMERISM**

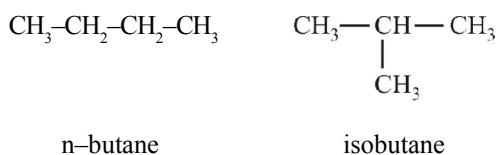
If the isomers have the same molecular formula but they differ in the relative arrangement of atoms, it is called structural isomerism. In structural isomers, the structural formula of the isomers differ whereas the molecular formula remains same. This type of isomerism is further divided into various types.

15.1 Chain or Nuclear Isomerism

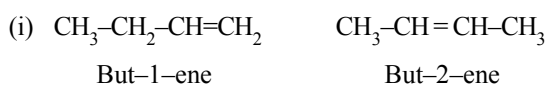
This type of isomerism arises due to the difference in the structure of carbon chain. The difference may be in the length of the carbon chain or in the size of the carbon ring.

Example 22

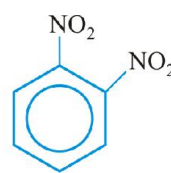
n-Butane and isobutane are chain isomers.

**15.2 Positional Isomerism**

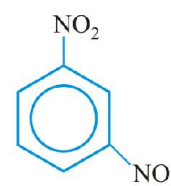
This isomerism arises due to the difference in position of either substituent or functional group in the same carbon chain.

Example 23

(ii) $\text{C}_6\text{H}_4(\text{NO}_2)_2$ exhibits following three positional isomers.



o-Dinitrobenzene



m-Dinitrobenzene



p-Dinitrobenzene

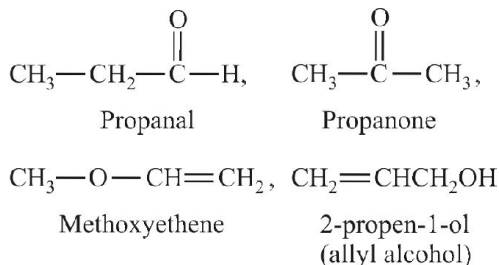
15.3 Functional Isomerism

These isomers have same molecular formula but they differ only in the presence of different functional groups.

Molecular Formula	Functional Groups
$\text{C}_n\text{H}_{2n+2}$	Alkane
C_nH_{2n}	Alkene, Cycloalkane
$\text{C}_n\text{H}_{2n-2}$	Alkyne, Cycloalkene, Diene (Cumulated, Conjugated, Isolated)
$\text{C}_n\text{H}_{2n+2}\text{O}$	Alcohols, Ethers
$\text{C}_n\text{H}_{2n}\text{O}$	Aldehydes, Ketones, Cyclic Ether (Epoxide, Oxirane), Unsaturated Alcohol
$\text{C}_n\text{H}_{2n}\text{O}_2$	Acid, Ester, Hydroxyaldehyde and Hydroxy Ketone

Example 24

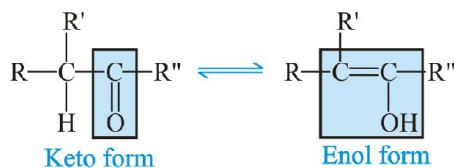
- (i) $\text{CH}_3-\text{C}\equiv\text{N}$ $\text{CH}_3-\text{N}^+\equiv\text{C}^-$
Methylcyanide Methylisocyanide
- (ii) $\text{C}_3\text{H}_6\text{O}$ exhibits the following functional isomers.



15.4 Tautomerism

This is a special type of functional isomerism where functional isomers exist in equilibrium with each other. Such isomers are called tautomers. The necessary condition for this type of isomerism is the presence of α -hydrogen to the carbonyl group (a hydrogen on a carbon adjacent to carbon of functional group).

A very common form of tautomerism is that between a carbonyl compound containing an hydrogen and its enol form. This type of isomerism is also known as keto-enol isomerism.

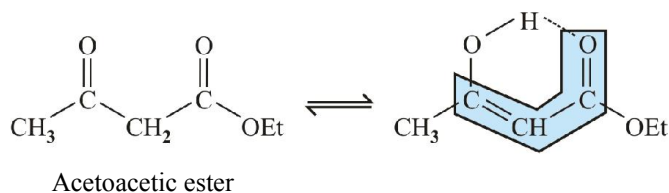


In general, the **keto form is more stable than the enol form** but in some cases the enol form is more stable.

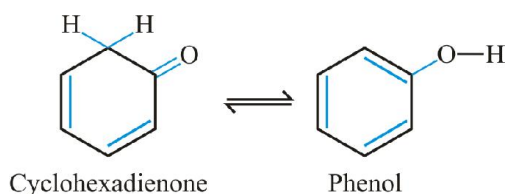
Cases when the enol form is more stable than the keto form :

- (1) Enol form has resonance stabilization by forming a conjugated system and has intra-molecular hydrogen bonding.

Example 25



- (2) Enol form has aromatic stabilization.



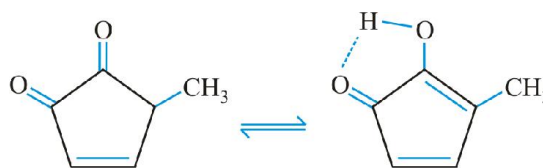
Note... Test for Enols

In those equilibria where the enol content is higher than the keto form, the solution produces red/green/blue colour with neutral FeCl_3 solution. The appearance of such a colour on addition of neutral FeCl_3 solution is a test

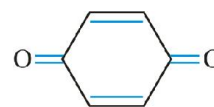
Example 26



Example 27



Example 28



This cannot show tautomerism as no sp^3 hybridized α -carbon is present.

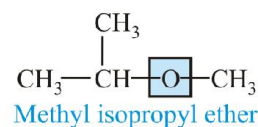
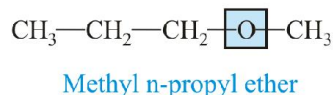
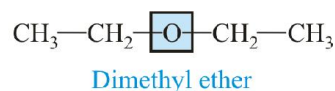
Note... The tautomers are not counted while finding the total number of isomers for a given molecular formula.

15.5 Metamerism

It is caused by the attachment of different alkyl/groups to a polyvalent atom or group (an atom having more than one valency). A metamer can be obtained by shifting one or more CH_2 group from one side of the polyvalent functional group to the other side. Metamerism is found to occur in amines, ethers, ketones, esters, etc.

Example 29

$\text{C}_4\text{H}_{10}\text{O}$ exhibits three metamers.



16. STEREO ISOMERISM

In this type of isomerism, the compounds possessing same molecular formula differ in their properties due to the difference in the arrangement in space of their atoms or groups in the molecule.

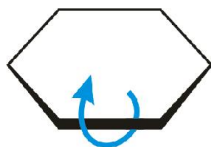
16.1 Geometrical Isomerism

It is the type of isomerism in which the compounds possessing same molecular formula differ in their properties due to the difference in their geometry that is, due to the difference in the direction of attachment of same atoms or groups in their molecule. It is not shown by single bonded compounds like (C–C) due to free rotation.

It is shown by $C=C$, $C=N$, $N=N$ and cycloalkanes.

Conditions for exhibiting Geometrical Isomerism

- (a) There must be restricted rotation in the molecule.



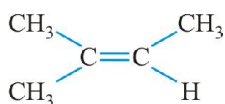
- (b) Groups attached to the doubly bonded carbon or nitrogen must be different. Lone pair is also considered as a different group.

16.1.1 Cis-Trans Isomerism

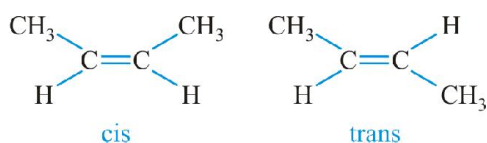
Geometrical Isomerism in $C=C$

Example 30

Same atoms or groups on same side. So geometrical isomerism is not possible.



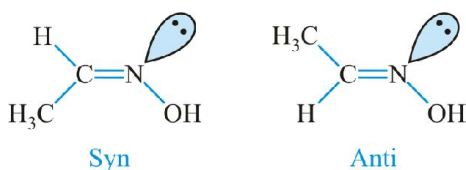
Example 31



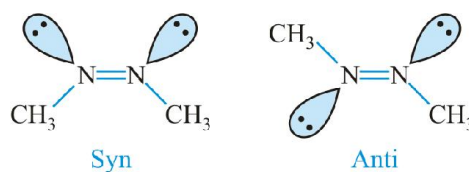
16.1.2 Syn-Anti Isomerism

Geometrical isomerism in $C=N$ and $N=N$

Example 32



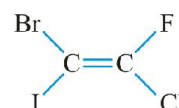
Example 33



16.1.3 E-Z Nomenclature of Geometrical isomers

When $C=C$ contain three or four different groups, it is difficult to assign cis or trans configuration to the geometrical isomers.

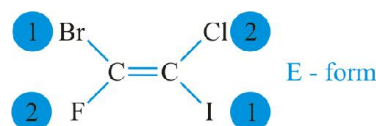
Example 34



To deal with such problems, E-Z nomenclature is used. E (similar to anti), Z (similar to syn)

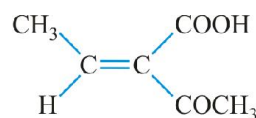
Cahn - Ingold - Prelog (CIP) Rules

- Higher atomic number group is assigned higher priority.

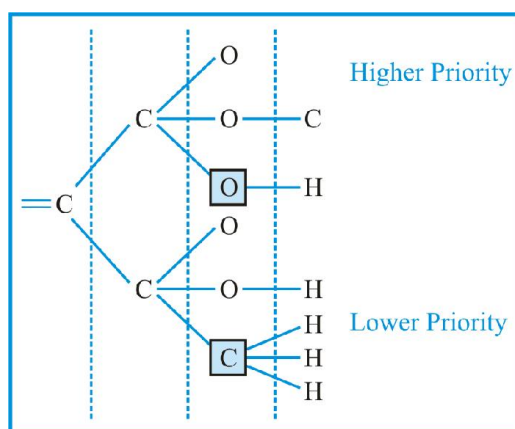


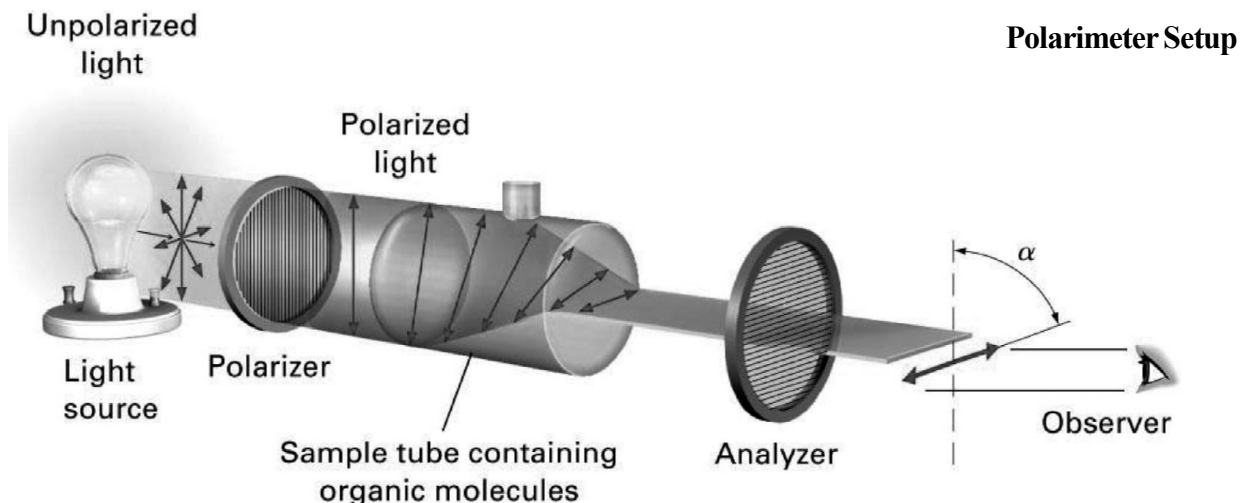
- In case of isotopes, check the atomic mass of that atom.
Example : H, D
- Chain Method

Example 35



For determining priority, use the following method





16.2 Optical Isomerism

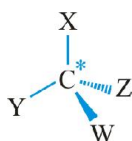
Optical isomers differ in their interaction with the plane polarized light. The property of rotating the plane polarized light is called **optical activity** and the compound that rotates the plane of polarized light is called optically active compound. The angle by which plane polarized light is rotated is called the **specific rotation** of the compound and is measured by **polarimeter**.

16.2.1 Chirality

Chirality means unsymmetricalness. A molecule that is non-superimposable on its mirror image is **chiral**. A molecule that is superimposable on its mirror image is **achiral**. Chirality is the necessary and sufficient condition for a compound to show optical activity.

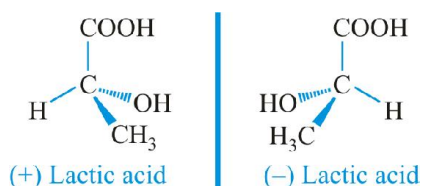
16.2.2 Chiral Carbon and Chiral Compound

When a carbon atom is bonded to four different atoms (or groups) CWXYZ, it is chiral and is denoted as C*.



A compound which is non-superimposable on its mirror image is called a chiral compound.

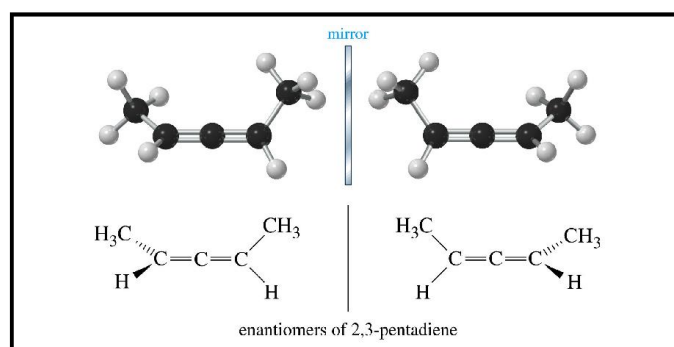
Example 36



Note...

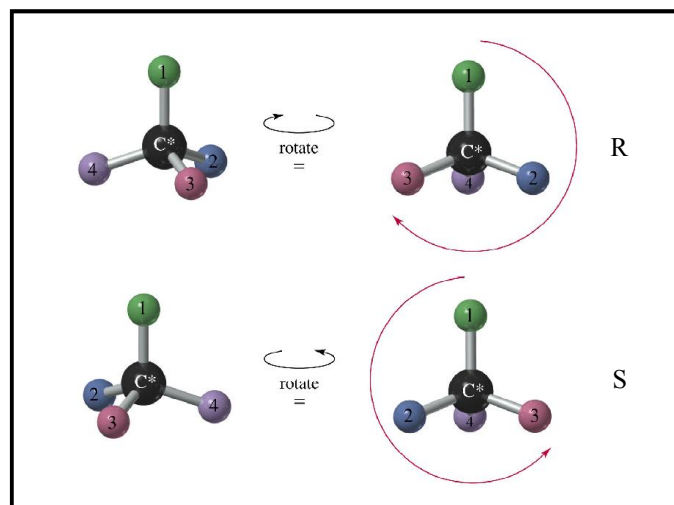
It is not necessary for a chiral compound to contain chiral carbon.

Example 37



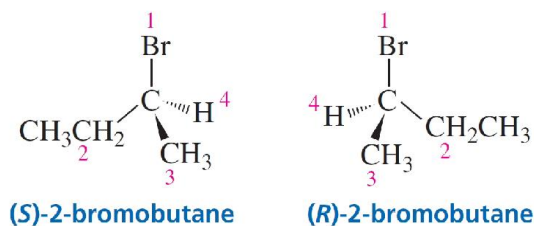
16.2.3 RS Configuration

The arrangement of atoms that characterizes a particular stereoisomer is called its configuration.

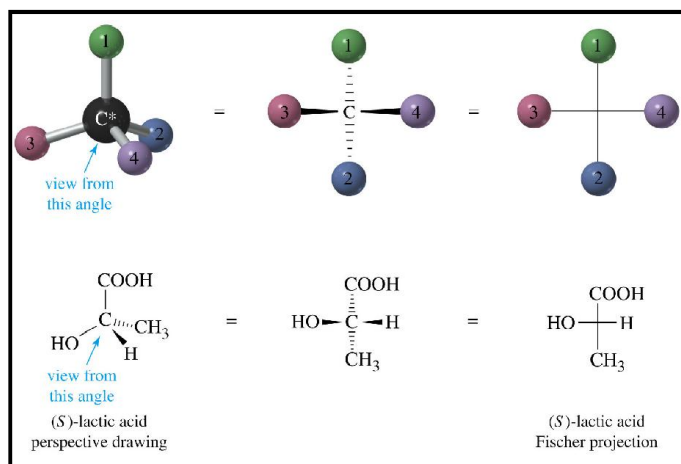


The configuration is assigned for a stereoisomer based on Cahn-Ingold-Prelog rules by assigning priority to the groups attached to the chiral carbon. The rules that are used for EZ isomerism are used in the same way here as well. The lowest priority group should be behind the plane while deciding the configuration.

If the direction of priorities assigned to groups is clockwise, the compound has R configuration. If the direction of priorities assigned to groups is anti-clockwise, the compound has S configuration.

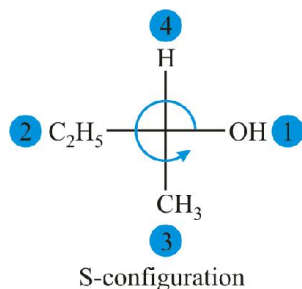
Example 38**16.2.4 Fischer Projection**

It is the two dimensional representation of a tetrahedral carbon. The method to draw Fischer projection of a molecule has been illustrated.

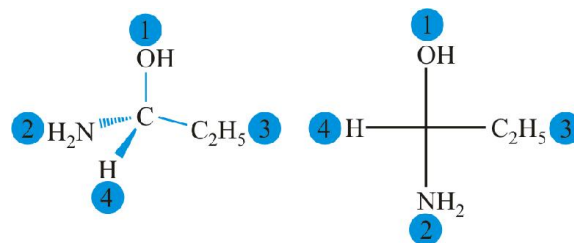


The groups present on the vertical line in Fischer projection are behind the plane whereas the groups present on the horizontal line are outside the plane.

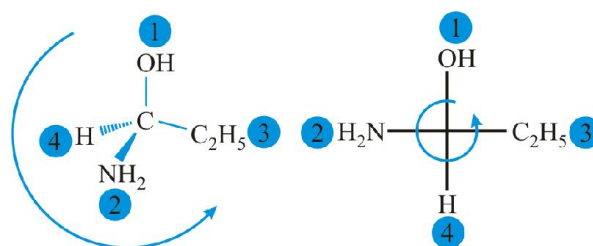
RS configuration is assigned in a Fischer projection by keeping the lowest priority group on the vertical line (behind the plane).

Example 39

If the lowest priority group is not behind the plane or on the vertical, exchange it with the group that is behind the plane. When two groups are exchanged, the configuration of that atom is inverted.

Example 40

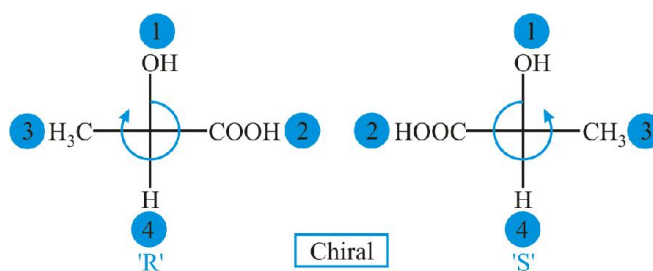
Exchange H and NH_2 and we get :



Therefore, after the exchange, we get S configuration which means the given molecule has R configuration.

16.2.5 Chiral Compounds with Chiral Carbons

Case I : Compounds with only one chiral carbon are always chiral, i.e. optically active.

Example 41

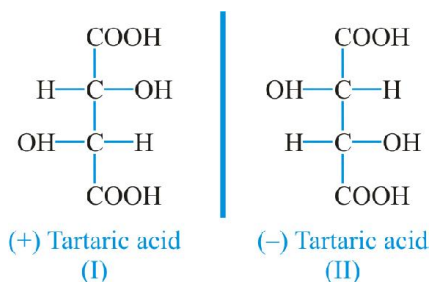
Note...

R and S configurations are mirror images of each other.

Case II : Chiral compounds containing more than one chiral carbon are chiral only if they do not satisfy any symmetry condition such as plane of symmetry or centre of symmetry. If any symmetrical condition is satisfied, the compound is achiral.

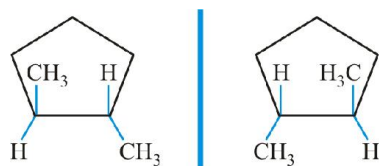
Example 42

Chiral



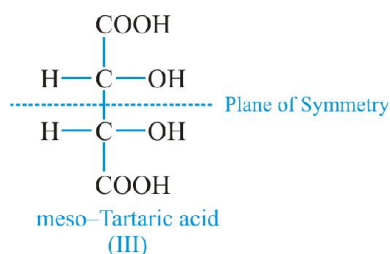
Example 43

Chiral



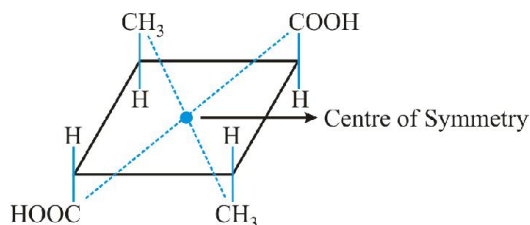
Example 44

Achiral



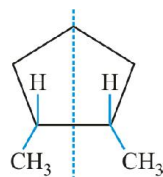
Example 45

Achiral



Example 46

Achiral



16.2.6 Enantiomers, Diastereomers, Meso Compounds and Racemic Mixture

Enantiomers are compounds that are non-superimposable mirror images of each other.

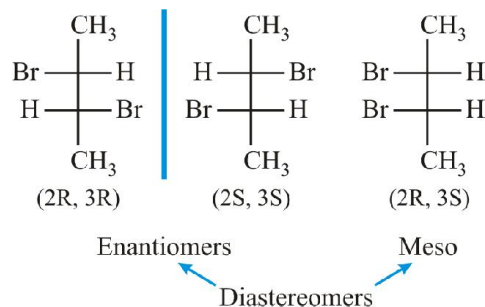
Meso compounds are compounds which satisfy any symmetry condition will be optically inactive due to internal compensation of rotation.

Diastereomers are stereoisomers which are non-mirror images of each other.

Racemic Mixture are equimolar mixture of enantiomers. They are optically inactive due to external compensation of rotation.

Resolution is the process of separation of racemic mixture into optically active enantiomers.

Example 47



16.2.7 Specific Rotation and Nature of Rotation

Optical activity is the ability to rotate the plane-polarized light. The optically active isomers are of two types

- Dextrorotatory** : The isomer that rotates the plane-polarized light to the right (clockwise) is called 'Dextrorotatory'. It is denoted as 'd' or (+).
- Laevorotatory** : The isomer that rotates the plane-polarized light to the left (anticlockwise) is called 'Laevorotatory'. It is denoted as 'l' or (-).

Optical rotation is measured by polarimeter in terms of specific rotation.

$$[\alpha]_D^{25} = \frac{\alpha_{\text{observed}}}{\ell \times c}$$

α = specific rotation

α_{observed} = observed value of rotation

ℓ = length of solution

c = concentration of solution.

Specific Rotation depends upon :

- Nature of solute and solvent.
- Wavelength of light during the experiment.
- Temperature during the experiment.
- Length and concentration of solution.



The *dℓ* configuration is relative to the optical activity experiment whereas *RS* configuration is absolute configuration which is assigned based on priority of groups according to CIP rules.

16.3 Conformers

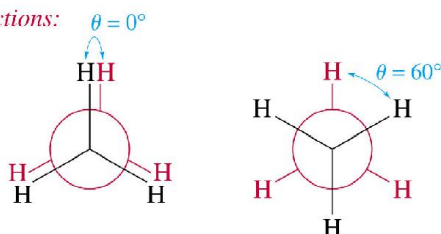
The different spatial arrangements of the atoms that result from rotation about a single bond are called conformations. A specific conformation is called a conformer.

When rotation occurs about the carbon-carbon bond of ethane, two extreme conformations can result—a staggered conformation and an eclipsed conformation. An infinite number of conformations between these two extremes are also possible.

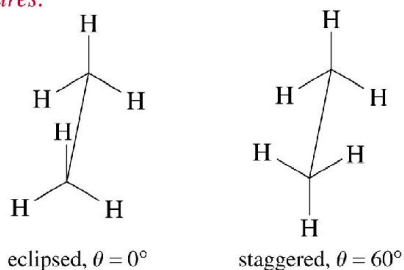
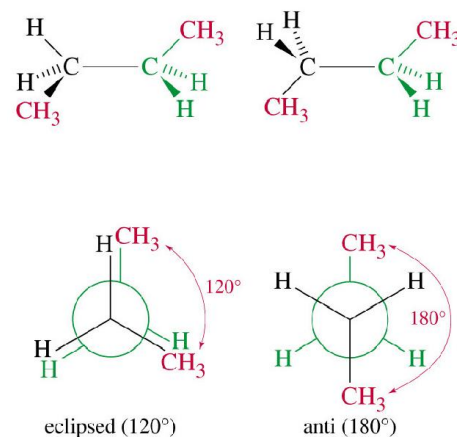
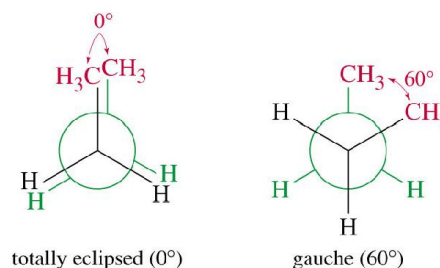
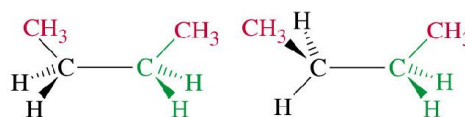
The staggered conformation, therefore, is the most stable conformation of ethane because the C–H bonds are as far away from each other as possible. The eclipsed conformation is the least stable conformation because in no other conformation are the C–H bonds as close to one another. The extra energy of the eclipsed conformation is called torsional strain.

16.3.1 Conformations of Ethane

Newman projections:



Sawhorse structures:

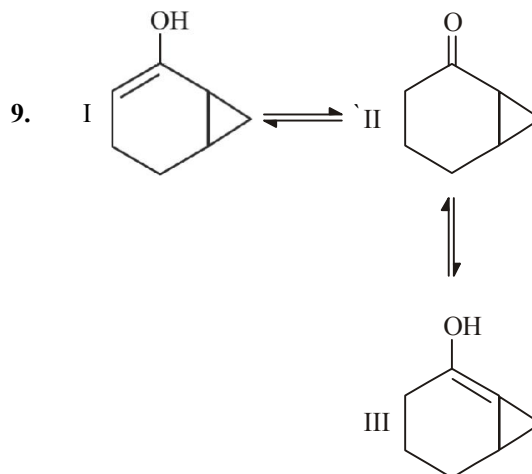
**16.3.2 Conformations of Butane**

Order of stability of conformers of Butane :

Anti > Gauche > Eclipsed > Totally Eclipsed

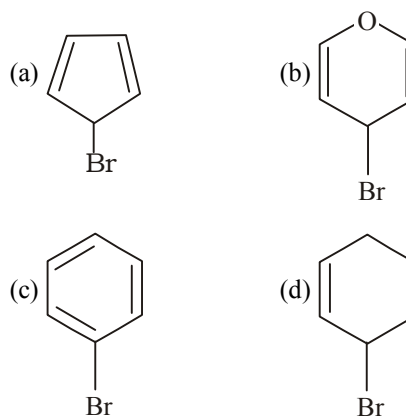
EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS**Revision : Isomers**

- How many isomers are possible with trisubstituted benzene with all the three groups same ?
(a) 3 (b) 4
(c) 5 (d) 6
- Which of the following is an example of functional isomerism ?
(a) Isopentane and Neopentane
(b) Glucose and Fructose
(c) Diethyl Ether and 2-Methoxy Propane
(d) α -Naphthol and β -Naphthol
- The possible number of alkynes with the formula C_5H_8 is
(a) 2 (b) 3
(c) 4 (d) 5
- The number of isomers for the aromatic compound C_7H_8O (containing benzene ring) are
(a) 2 (b) 3
(c) 4 (d) 5
- Rotation of polarised light can be measured by
(a) Monometer (b) Galvanometer
(c) Polarimeter (d) Viscometer
- The most stable conformation of ethylene glycol is
(a) anti (b) gauche
(c) fully eclipsed (d) partially eclipsed
- The unsaturation factor of the molecular formula C_5H_8O is
(a) 1 (b) 2
(c) 3 (d) 4
- In unsaturation factor is two, then it means that the structure should have
(a) either two π -bonds or two rings or one ring with a π -bonds.
(b) only two π -bonds
(c) only two rings
(d) both two π -bonds and two rings



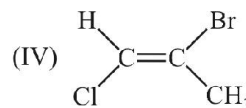
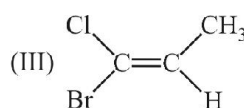
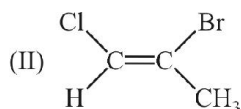
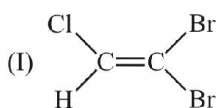
Stability order among these tautomers is

- (a) $I > II > III$ (b) $III > II > I$
(c) $II > I > III$ (d) $II > III > I$
10. Ease of ionization to produce carbocation and bromide ion under the treatment of Ag^+ will be maximum in which of the following compounds?

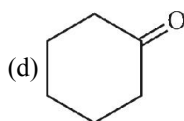
**Geometrical Isomers**

11. Of the following compounds, which will have zero dipole moment ?
(a) 1, 1-Dichloroethylene
(b) cis-1, 2-Dichloroethylene
(c) trans-1, 2-Dichloroethylene
(d) None of these compounds

12. Maleic acid and fumaric acid are
(a) Position isomers (b) Geometrical isomers
(c) Enantiomers (d) Functional isomers
13. Geometrical isomerism will be exhibited by
(a) Pent-1-ene
(b) 3-Methylbut-1-ene
(c) Pent-2-ene
(d) All of these compounds
14. Which of the following compounds can show geometrical isomerism?
(a) Vinyl chloride (b) 1, 1-Dichloroethene
(c) 1, 2-Dichloroethene (d) Trichloroethene
15. The simplest alkene which is capable of exhibiting geometrical isomerism has
(a) 3 carbon atoms (b) 5 carbon atoms
(c) 6 carbon atoms (d) 4 carbon atoms
16. Which of the following compounds does not exhibit geometrical isomerism?
(a) 1, 1-Dichlorobut-2-ene
(b) 1, 2-Dichlorobut-2-ene
(c) 1, 1-Dichlorobut-1-ene
(d) 2, 3-Dichlorobut-2-ene
17. Which of the following compounds can exhibit geometrical isomerism?
(a) Hex-1-ene (b) 2-Methylpent-2-ene
(c) 3-Methylpent-1-ene (d) Hex-2-ene
18. Each of the following compounds show geometrical isomerism except
(a) But-2-en-1, 4-dioic acid
(b) But-2-yn-1, 4-dioic acid
(c) But-2-ene
(d) 3-Chlorohex-2-ene
19. Which of the following can exist as cis and trans isomers?
(a) $\text{CH}_2=\text{CH}-\text{Cl}$ (b) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$
(c) $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCl}$ (d) $\text{CHCl}=\text{CCl}_2$
20. Which is a pair of geometrical isomers?



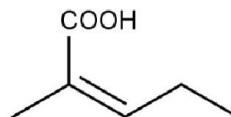
- (a) I and II (b) I and III
(c) II and IV (d) III and IV
21. Which of the following compound will not exhibit geometrical isomerism?
(a) $\text{CH}_3-\text{CH}=\text{CH}-\text{COOH}$
(b) $\text{Br}-\text{CH}=\text{CH}-\text{Br}$
(c) $\text{C}_6\text{H}_5-\text{CH}=\text{NOH}$



E/Z Isomers

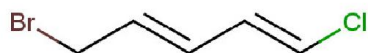
22. The correct configuration name of
- (a) (2E, 5E) (b) (2Z, 5Z)
(c) (2E, 5Z) (d) (2Z, 5E)

- 23.
- This compound can be named as
(a) only cis-1-bromo-1-chloro-propene
(b) only trans-1-bromo-1-chloro-propene
(c) trans-1-bromo-1-chloro-propene as well (z)-1-bromo-1-chloro-propene
(d) only (E)-1-bromo-1-chloropropene
24. What is the IUPAC name of the following compound?



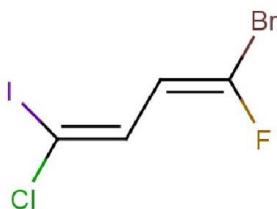
- (a) (Z)-2-Methylpent-2-enoic acid
(b) (E)-2-Methylpent-2-enoic acid
(c) (Z)-2-Methylpent-2-ene carboxylic acid
(d) (E)-2-Methylpent-2-ene carboxylic acid

25. What is the configuration of the given molecule ?



- (a) (1E,3Z) (b) (1E,3E)
(c) (1Z,3Z) (d) (1Z,3E)

26. What is the IUPAC name of the following compound ?



- (a) (1Z,3E)-4-Bromo-1-chloro-4-fluoro-1-iodobuta-1,3-diene
(b) (1E,3Z)-4-Bromo-1-chloro-4-fluoro-1-iodobuta-1,3-diene
(c) (1Z,3E)-1-Bromo-4-chloro-1-fluoro-4-iodobuta-1,3-diene
(d) (1E,3Z)-1-Bromo-4-chloro-1-fluoro-4-iodobuta-1,3-diene

Stereocentres

27. Which of following have asymmetric carbon atom ?

- (a) $\text{CH}_2\text{Cl}-\text{CH}_2\text{Br}$ (b) CH_3CHCl_2
(c) $\text{CH}_3\text{CH}_2\text{Cl}$ (d) $\text{CH}_2\text{Br}-\text{CHOH}-\text{CH}_3$

28. How many asymmetric carbon atoms are present in :

- (i) 1, 2-dimethyl cyclohexane
(ii) 3-methyl cyclopentene
(iii) 3-methyl cyclohexene

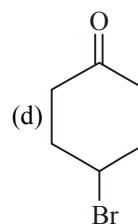
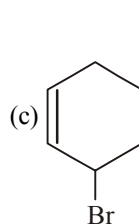
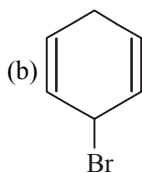
- (a) two, one, one (b) one, one, one
(c) two, none, two (d) two, none, one

29. A stereo-center is

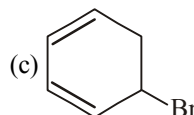
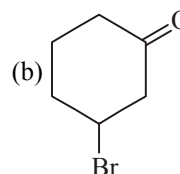
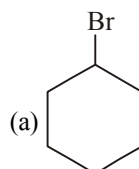
- (a) An atom in which an interchange of any two group will result in a new stereoisomer
(b) A chiral carbon atom
(c) Both of these
(d) none of these

30. Which of the following compounds has asymmetric centre ?

- (a) $\text{CH}_2=\text{CH}-\text{CH}(\text{Br})-\text{CH}=\text{CH}_2$

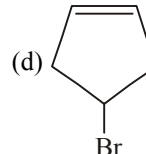
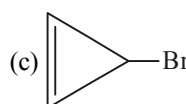
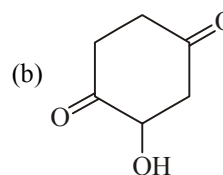
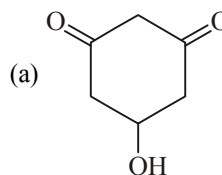


31. Which of the following compounds has asymmetric centre ?

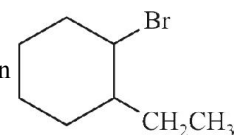


- (d) both (b) and (c)

32. Which of the following compounds has asymmetric centre ?



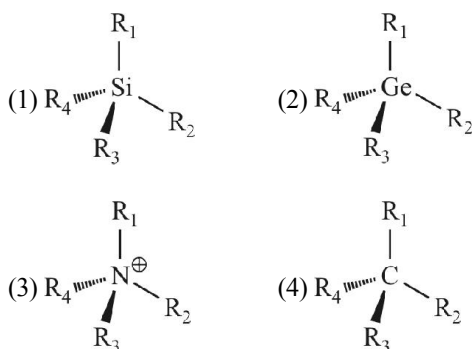
33. Number of chiral carbon atom in



- (a) 2 (b) 4
(c) 3 (d) 1

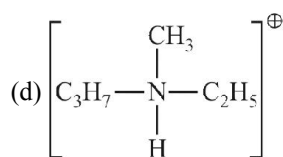
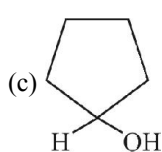
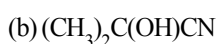
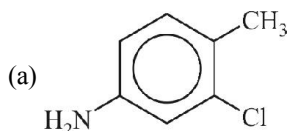
Chiral Compounds

34. Consider following structures and write correct statements

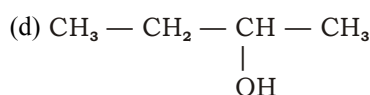
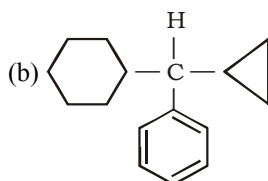
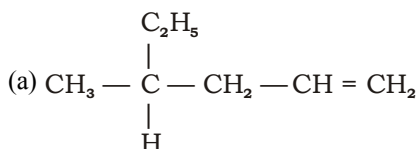


where $R_1 \neq R_2 \neq R_3 \neq R_4$

- (a) Only 4 shows optical isomerism
 (b) 1 and 2 show optical isomerism
 (c) 1, 2 and 4 show optical isomerism
 (d) All show optical isomerism
35. Consider the following organic compound.
 $\overset{1}{\text{CH}_3}-\overset{2}{\text{CH}_2}-\overset{3}{\text{CH}_2}-\overset{4}{\text{CH}_2}-\overset{5}{\text{CH}_2}-\overset{6}{\text{CH}_2}-\overset{7}{\text{CH}_3}$
 To make it a chiral compound, the attack should be on carbon
 (a) 1 (b) 3
 (c) 4 (d) 7
36. Which of the following compound may exhibit optical activity?



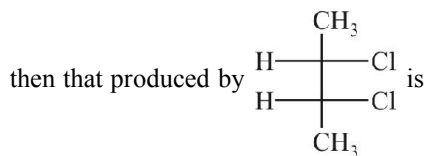
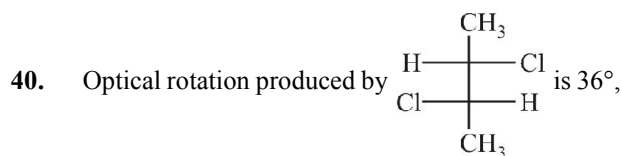
37. Which of the following will not show optical activity?



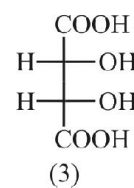
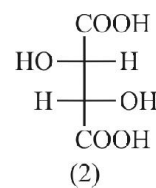
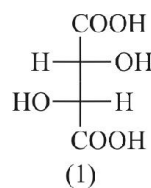
38. An organic compound $\text{C}_4\text{H}_8\text{O}$, is found to be optically active. Which of the following could it be?
- (a) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CHO}$
 (b) $\text{CH}_2=\text{CO}-\text{CH}_2-\text{CH}_3$
 (c) $\text{CH}_2=\text{CH}-\text{CH}(\text{OH})-\text{CH}_3$
 (d) $(\text{CH}_3)_2\text{CH}-\text{CHO}$

Optical Activity

39. Rotation of polarised light can be measured by
 (a) Monometer (b) Galvanometer
 (c) Polarimeter (d) Viscometer



- (a) -36° (b) 0°
 (c) $+36^\circ$ (d) unpredictable
41. From the structure of the three stereoisomers of tartaric acid, which statement should be true?



- (a) 1 is (+), 2 is (-), and 3 is optically inactive
 (b) 1 is (-), 2 is (+), and 3 is optically inactive
 (c) 1 is (+), 2 is (-), and 3 is meso
 (d) 3 is optically inactive, but nothing can be said about 1 and 2

42. When an optically active compound is placed in a 10 dm tube, is present 20 g in a 200 mL solution rotates the plane polarized light by 30° . Calculate the angle of rotation and specific angle of rotation if above solution is diluted to 1 L.

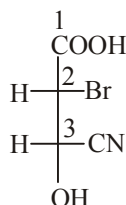
(a) 16° and 36° (b) 6° and 30°
(c) 3° and 30° (d) 6° and 36°

R/S Configuration

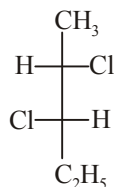
43. Rank of the following groups in decreasing order of R, S precedence.

$-\text{CH}(\text{CH}_3)_2$	$-\text{CH}_2\text{CH}_2\text{Br}$	$-\text{CH}_2\text{Br}$	$-\text{C}(\text{CH}_3)_3$
1	2	3	4
(a) 3	2	4	1
(b) 1	4	2	3
(c) 3	4	1	2
(d) 3	4	2	1

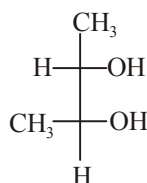
44. The absolute configurations of the two centres in the following molecules are:



- (a) 2(R), 3(S) (b) 2(R), 3(R)
(c) 2(S), 3(S) (d) 2(S), 3(R)
45. The configuration at C-2 and C-3 of the compound given:

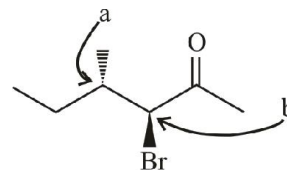


- (a) 2R, 3S (b) 2S, 3R
(c) 2S, 3S (d) 2R, 3R
46. Correct configuration of the following molecule is:

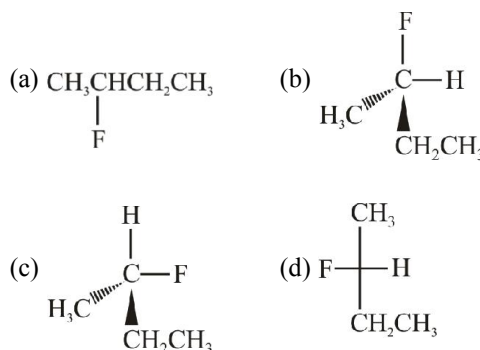


(a) 1S, 2S (b) 1S, 2R
(c) 1R, 2S (d) 1R, 2R

47. Determine the absolute configurations of the chiral centres in the following compound.

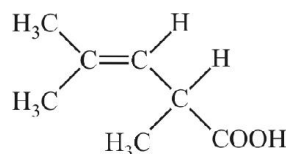


- (a) a=R; b=S (b) a=R; b=R
(c) a=S; b=S (d) a=S; b=R
48. The structure of (S)-2-fluorobutane is best represented by:

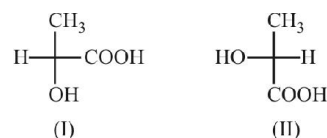


Stereoisomers

49. The following compound can exhibit :



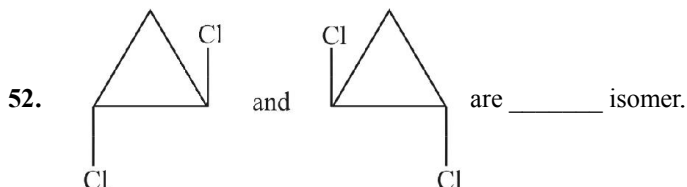
- (a) geometrical isomerism
(b) geometrical and optical isomerism
(c) optical isomerism
(d) tautomerism
50. The two structures (I) and (II) represented below are



- (a) enantiomers
(b) diastereomers
(c) two representations of the same molecule
(d) functional isomers

51. A pair of optically active compounds which are not the mirror images are called

- (a) Mesomers (b) Anomers
(c) Epimers (d) Diastereomers

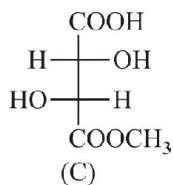
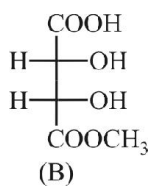
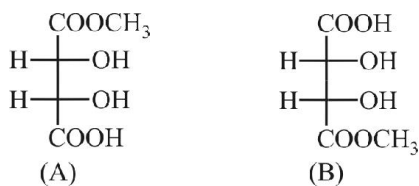


- (a) Position (b) Optical
(c) Geometrical (d) Diastereomers

53. Meso form of tartaric acid is :

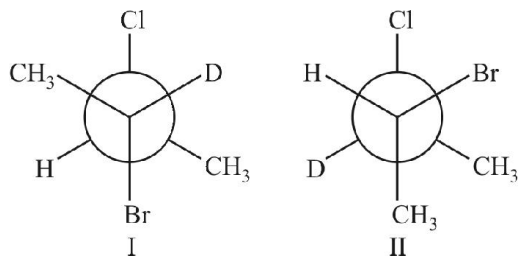
- (a) Dextrorotatory (b) Laevorotatory
(c) Neither Laevo nor dextro rotatory due to internal compensation
(d) A mixture of equal quantities of dextro and levo rotatory forms

54. The correct statement about the compound A, B and C



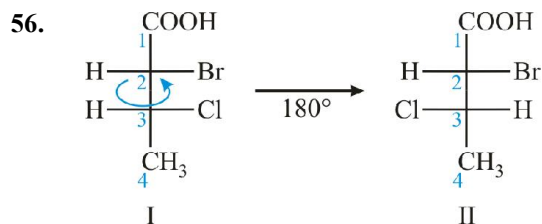
- (a) A and B are identical
(b) A and B are diastereomers
(c) A and C are enantiomers
(d) A and B are enantiomers

55. Which of the following statements is true about these two structures ?



- (a) I and II are a pair of enantiomers

- (b) I and II are a pair of diastereomers
(c) I and II are a pair of conformers
(d) I and II are a pair of structural isomers

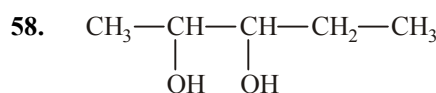


Holding C-2 unit of I in hand, rotation of C-3 unit about C2-C3 bond by 180° seems to produce II. Which of the following statements is correct ?

- (a) I and II are a pair of enantiomers
(b) I and II are a pair of diastereomers
(c) I and II are a pair of conformers
(d) I and II are identical

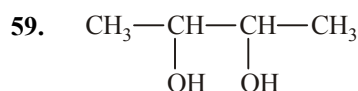
57. Number of stereoisomers of the compound, 2-chloro-4-methylhex-2-ene is

- (a) 2 (b) 4
(c) 8 (d) 16



How many optical isomers are possible in this compound ?

- (a) 2 (b) 3
(c) 4 (d) 5

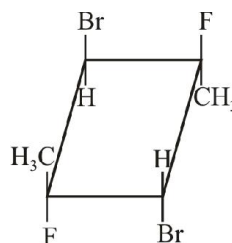


How many optical isomers are possible in this compound ?

- (a) 2 (b) 3
(c) 4 (d) 5

Chirality in Cyclic Compounds

60. The compound has:

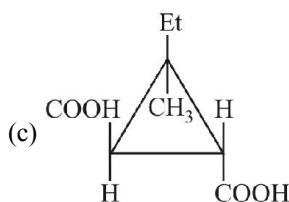
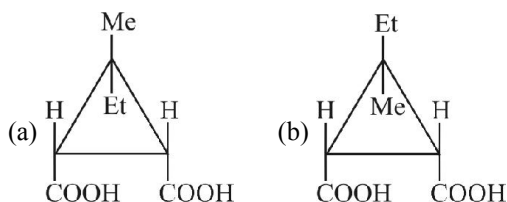


- (a) plane of symmetry (b) axis of symmetry
(c) center of symmetry (d) no symmetry

61. The number of enantiomers of the compound $\text{CH}_3\text{CHBrCHBrCOOH}$ is

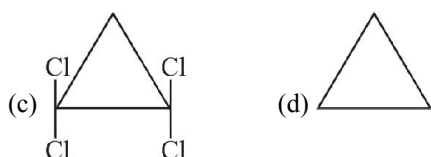
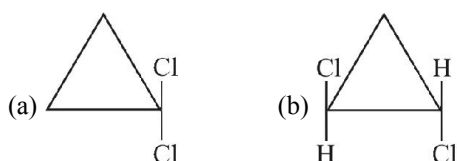
- (a) 1 (b) 2
(c) 3 (d) 4

62. Which of the following is optically active ?



(d) All of these

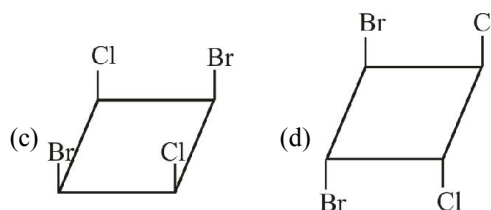
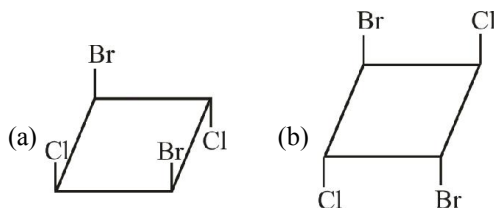
63. Which of the following is optically active ?



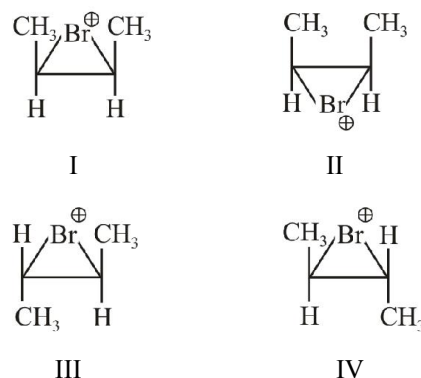
64. Which of the following shows optical isomerism ?

- (a) cis-1, 2-Dimethyl cyclohexane
(b) trans-1, 3-Dimethyl cyclohexane
(c) trans-1, 4-Dimethyl cyclohexane
(d) All of the above

65. Which of the following structures is chiral?



66.

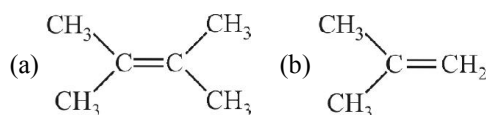


Which of the following statements is true about these molecules?

- (a) I is a mesomer
(b) I and II are identical
(c) III and IV are a pair of enantiomers
(d) all of these

Racemization

67. Which one of the following alkene, when reacted with HCl produces racemic mixture ?



(c) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$

(d) $\text{CH}_3\text{CH}=\text{CH}_2$

68. The conversion of dextro- or laevo- form into racemic mixture is called

- (a) Resolution (b) Racemisation
(c) Dehydration (d) Revolution

Resolution

69. An enantiomerically pure acid is treated with racemic mixture of an alcohol having one chiral carbon. The ester formed will be

- (a) Optically active mixture
(b) Pure enantiomer
(c) Meso compound
(d) Racemic mixture

70. The process of separation of racemic mixture into d and l enantiomers is called

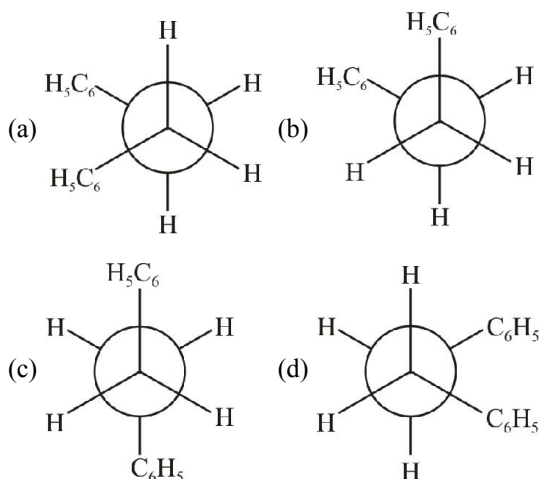
- (a) resolution (b) racemisation
(c) filtration (d) distillation

Conformers

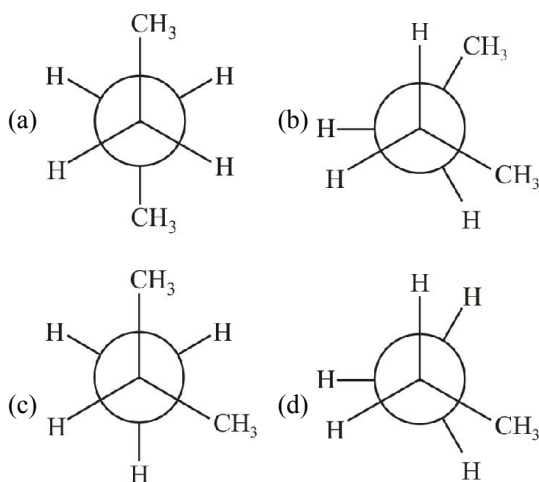
71. The Baeyer's angle strain is expected to be maximum in

- (a) cyclopentane (b) cyclodecane
(c) cyclooctane (d) cyclohexane

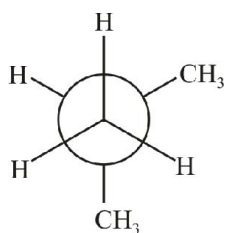
72. The most stable conformation of 1,2-diphenylethane is



73. Which of the following is a conformer of isobutane?



74. The Newman projection below represents



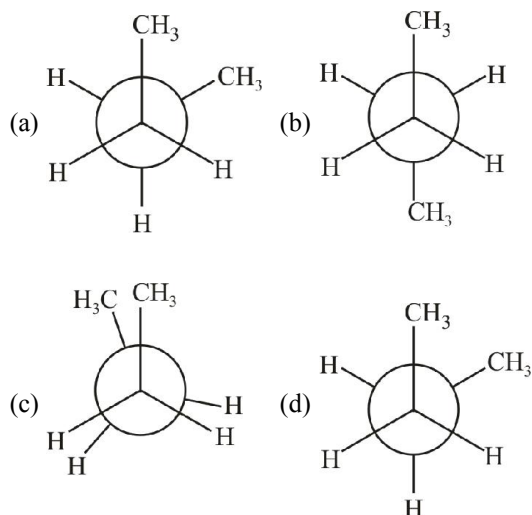
(a) A staggered form of 2-methyl propane

(b) A gauche form of butane

(c) An eclipsed form of 2-methyl propane

(d) A most stable conformation of 2-methyl butane

75. Which conformation of the alkane has the highest potential energy?



Total Number of Isomers

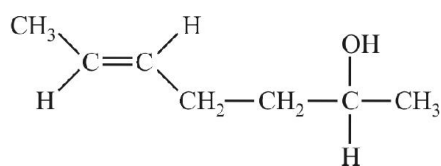
76. The molecular formula of a saturated compound is $C_2H_4Cl_2$. The formula permits the existence of two

- (a) functional isomers (b) position isomers (c) optical isomers
(d) cis-trans isomers

77. Only two isomeric monochloro derivative are possible for

- (a) n-Butane (b) 2, 4-Dimethylpentane
(c) Benzene (d) 2-Methyl propane

78. The compound, whose stereochemical formula is written below, exhibits x geometrical isomers and y optical isomers



The values of x and y are :

- (a) 4 and 4 (b) 2 and 2
(c) 2 and 4 (d) 4 and 2

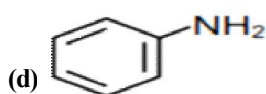
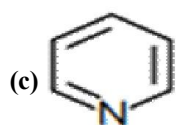
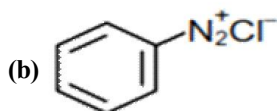
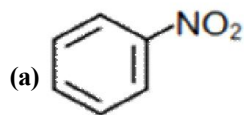
79. The number of geometrical isomers of the given compound $CH_3-CH=CH-CH=CH-C_2H_5$ is

- (a) 2 (b) 3
(c) 4 (d) 5

80. The number of isomers of dibromo derivative of an alkene (molar mass 186 g mol^{-1}) is
(a) two (b) three
(c) four (d) six
81. The number of optical active isomers possible for $\text{CH}_3\text{CH}=\text{CH}=\text{CHCH}_3$ is
(a) 2 (b) 3
(c) 6 (d) 8
82. $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3 \xrightarrow[\text{H}_2\text{O}]{\text{H}} \text{Alkene (F)}$
 $\xrightarrow{\text{Br}_2/\text{CCl}_4} \text{C}_4\text{H}_8\text{Br}_2$ (5 products are possible).
How many structures of F are possible ?
(a) 2 (b) 5
(c) 6 (d) 3
83. The number of stereoisomers for pent-3-en-2-ol is
(a) 2 (b) 3
(c) 4 (d) 5
84. $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}=\text{CH}_2$
How many geometrical and optical isomers are possible in this compound ?
(a) 0, 2 (b) 0, 3
(c) 2, 0 (d) 2, 2
85. Number of Stereoisomers for $\text{CHO}-\text{CHOH}-\text{CHOH}-\text{CH}_2\text{OH}$ is
(a) 2 (b) 4
(c) 6 (d) 8

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTIONS

1. Which of the following compounds will be suitable for Kjeldahl's method for nitrogen estimation ? (2017)



2. Which of the following statements is not true about partition chromatography ? (2016)

- (a) Mobile phase can be a gas.
(b) Separation depends upon equilibration of solute between a mobile and a stationary phase.
(c) Paper chromatography is an example of partition chromatography.
(d) Stationary phase is a finely divided solid absorbent.

3. The optically inactive compound from the following is (2015)

- (a) 2-chloropropanal (b) 2-chloropentane
(c) 2-chlorobutane (d) 2-chloro-2-methylbutane

4. Which one of the following acids does not exhibit optical isomerism ? (2014)

- (a) Lactic acid (b) Tartaric acid
(c) Maleic acid (d) α -amino acids

5. Identify the compound that exhibits tautomerism (2011)

- (a) 2-butene (b) lactic acid
(c) 2-pentanone (d) phenol

6. Out of the following, the alkene that exhibits optical isomerism is (2010)

- (a) 3-methyl-2-pentene (b) 4-methyl-1-pentene
(c) 3-methyl-1-pentene (d) 2-methyl-2-pentene

7. The alkene that exhibits geometrical isomerism is (2009)

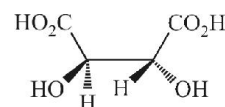
- (a) propene (b) 2-methyl propene
(c) 2-butene (d) 2-methyl-2-butene

8. The number of stereoisomers possible for a compound of the molecular formula



- (a) 3 (b) 2
(c) 4 (d) 6

9. The absolute configuration of (2008)

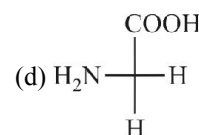
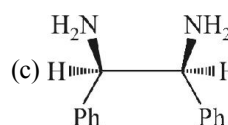
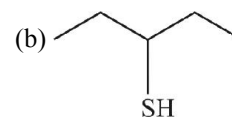
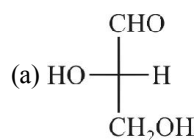


- (a) S, S (b) R, R
(c) R, S (d) S, R

10. Which one of the following conformations of cyclohexane is chiral ? (2007)

- (a) Twist boat (b) Rigid
(c) Chair (d) Boat

11. Which of the following molecules is expected to rotate the plane of polarised light ? (2007)



12. Increasing order of stability among the three main conformations (Eclipse, Anti, Gauche) of 2-fluoroethanol is (2006)

- (a) Eclipse, Gauche, Anti (b) Gauche, Eclipse, Anti
(c) Eclipse, Anti, Gauche (d) Anti, Gauche, Eclipse

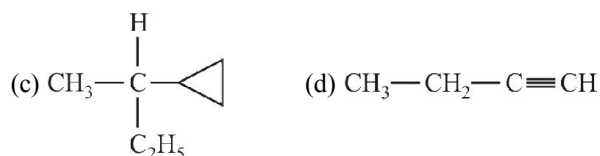
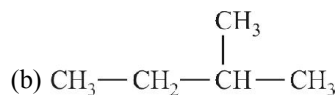
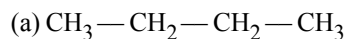
13. Which types of isomerism is shown by 2,3-dichlorobutane ? (2005)

- (a) Structural (b) Geometric
(c) Optical (d) Diastereomerism

14. Which of the following compounds is not chiral ? (2004)

- (a) 1-Chloropentane (b) 2-Chloropentane
(c) 1-Chloro-2-Methyl Pentane
(d) 3-Chloro-2-Methyl Pentane

15. Amongst the following compounds, the optically active alkane having lowest molecular mass is (2004)



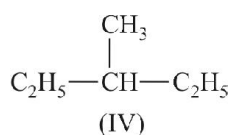
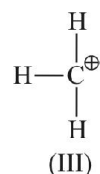
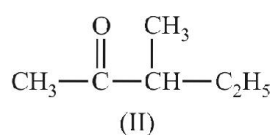
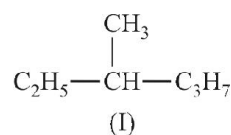
16. Which of the following will have a meso-isomer also ? (2004)

- (a) 2-Chlorobutane (b) 2, 3-Dichlorobutane
(c) 2, 3-Dichloropentane
(d) 2-Hydroxypropanoic Acid

17. For which of the following parameters the structural isomers $\text{C}_2\text{H}_5\text{OH}$ and CH_3OCH_3 would be expected to have the same values ? (Assume ideal behaviour) (2004)

- (a) Heat of vaporisation
(b) Vapour pressure at the same temperature
(c) Boiling points
(d) Gaseous densities at the same temperature and pressure

18. Among the following four structures I to IV (2003)

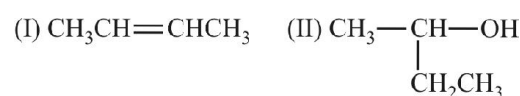


It is true that

- (a) all four are chiral compounds
(b) only I and II are chiral compounds

- (c) only III is a chiral compound
(d) only II and IV are chiral compounds

19. Following types of compounds (as I, II) (2002)



are studied in terms of isomerism in

- (a) chain isomerism (b) position isomerism
(c) conformers (d) stereoisomerism

20. Racemic mixture is formed by mixing two (2002)

- (a) isomeric compounds (b) chiral compounds
(c) meso compounds
(d) enantiomers with chiral carbon

21. Geometrical isomerism is not shown by (2002)

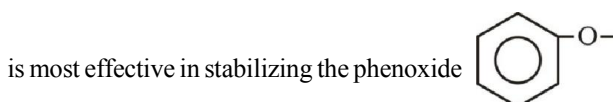
- (a) 1, 1-Dichloropent-1-ene
(b) 1, 2-Dichloropent-1-ene
(c) 1, 3-Dichloropent-2-ene
(d) 1, 4-Dichloropent-2-ene

JEE MAINS ONLINE QUESTION

1. Which one of the following acids does not exhibit optical isomerism? Online 2014 SET (3)

- (a) Maleic acid (b) Tartaric acid
(c) Lactic acid (d) α -amino acid

2. Which one of the following substituents at para-position



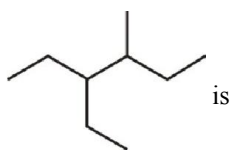
ions ? Online 2014 SET (4)

- (a) $-\text{OCH}_3$ (b) $-\text{CH}_2\text{OH}$
(c) $-\text{CH}_3$ (d) $-\text{COCH}_3$

3. Which one of the following compounds will not be soluble in sodium bicarbonate ? Online 2014 SET (4)

- (a) 2, 4, 6-Trinitrophenol
(b) Benzoic acid
(c) o-Nitrophenol
(d) Benzene sulphonic acid

4. The correct IUPAC name of the following compound :



is

Online 2014 SET (4)

- (a) 4-methyl-3-ethylhexane
(b) 3-ethyl-4-methylhexane
(c) 3,4-ethylmethylhexane
(d) 4-ethyl-3-methylhexane

5. The number of structural isomers for C_6H_{14} is :

Online 2015 SET (1)

- (a) 4 (b) 3
(c) 6 (d) 5

6. Match the organic compounds in column-I with the Lassaigne's test results in column-II appropriately :

Column-I

(A) Aniline

(B) Benzenesulfonic acid

(C) Thiourea

Column-II

(i) Red colour with



(ii) Violet colour with sodium nitroprusside

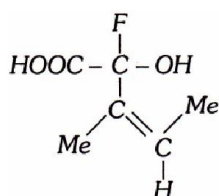
(iii) Blue colour with hot and acidic solution of $FeSO_4$

Online 2015 SET (1)

- (a) A-(ii); B-(iii); C-(i) (b) A-(iii); B-(i); C-(ii)
(c) A-(iii); B-(ii); C-(i) (d) A-(ii); B-(i); C-(iii)

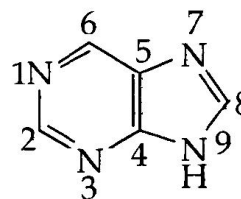
7. The configuration of the chiral centre and the geometry of the double bond in the following molecule can be described by

Online 2016 SET (1)



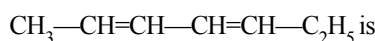
- (a) R and E (b) S and E
(c) R and Z (d) S and Z

8. The "N" which does **not** contribute to the basicity for the compound is :
Online 2016 SET (2)



- (a) N 7 (b) N 9
(c) N 1 (d) N 3

9. The number of geometrical isomers in the following compound



Online 2017 SET (1)

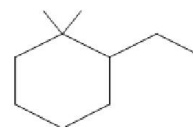
- (a) 4 (b) 3
(c) 2 (d) 5

10. Which of the following statements is **not** true about partition chromatography ?

Online 2017 SET (1)

- (a) Mobile phase can be a gas
(b) Stationary phase is a finely divided solid adsorbent
(c) Separation depends upon equilibration of solute between a mobile and a stationary phase
(d) Paper chromatography is an example of partition chromatography

11. The IUPAC name of the following compound is :

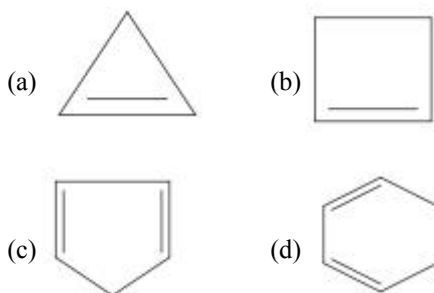


Online 2017 SET (1)

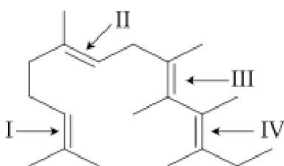
- (a) 1,1-Dimethyl-2-ethylcyclohexane
(b) 2-Ethyl-1,1-dimethylcyclohexane
(c) 1-Ethyl-2,2-dimethylcyclohexane
(d) 2,2-Dimethyl-1-ethylcyclohexane

12. Which of the following compounds is most reactive to an aqueous solution of sodium carbonate ?

Online 2017 SET (2)



13. In the following structure, the double bonds are marked as I, II, III and IV



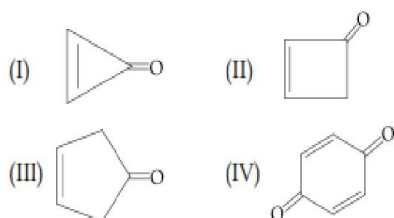
Geometrical isomerism is not possible at site (s) :

Online 2017 SET (2)

- (a) III (b) I
(c) I and III (d) III and IV

14. Which of the following compounds will show highest dipole moment

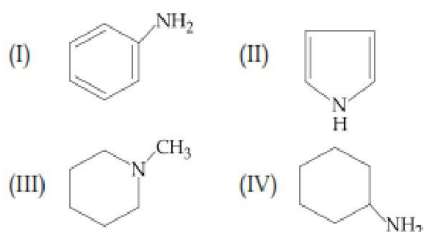
Online 2017 SET (2)



- (a) I (b) II
(c) III (d) IV

15. Among the following compounds, the increasing order of their basic strength is :

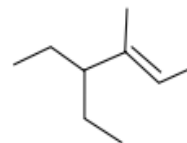
Online 2017 SET (2)



- (a) (I) < (II) < (IV) < (III) (b) (I) < (II) < (III) < (IV)
(c) (II) < (I) < (IV) < (III) (d) (II) < (I) < (III) < (IV)

16. The IUPAC name of the following compound is :

Online 2018 SET (1)



- (a) 4-methyl-3-ethylhex-4-ene
(b) 3-ethyl-4-methylhex-4-ene
(c) 4-ethyl-3-methylhex-2-ene
(d) 4, 4-diethyl-3-methylbut-2-ene

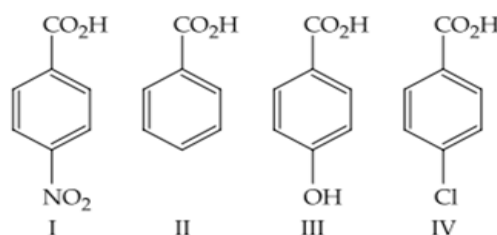
17. Two compounds I and II are eluted by column chromatography (adsorption of I > II). Which one of following is a correct statement ?

Online 2018 SET (2)

- (1) I moves faster and has higher R_f value than II
(2) II moves faster and has higher R_f value than I
(3) I moves slower and has higher R_f value than II
(4) II moves slower and has higher R_f value than I

18. The increasing order of the acidity of the following carboxylic acids is :

Online 2019 SET (1)



- (1) I < III < II < IV (2) IV < II < III < I
(3) II < IV < III < I (4) III < II < IV < I

19. In the following compounds, the decreasing order of basic strength will be :

Online 2019 SET (3)

- (a) $C_2H_5NH_2 > NH_3 > (C_2H_5)_2NH$
(b) $NH_3 > C_2H_5NH_2 > (C_2H_5)_2NH$
(c) $(C_2H_5)_2NH > C_2H_5NH_2 > NH_3$
(d) $(C_2H_5)_2NH > NH_3 > C_2H_5NH_2$

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

1. All questions marked “S” are single choice questions
2. All questions marked “M” are multiple choice questions
3. All questions marked “C” are comprehension based questions
4. All questions marked “A” are assertion–reason type questions
 - (A) If both assertion and reason are correct and reason is the correct explanation of assertion.
 - (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
 - (C) If assertion is true but reason is false.
 - (D) If reason is true but assertion is false.
5. All questions marked “X” are matrix–match type questions
6. All questions marked “I” are integer type questions

Revision : Isomers

1. (A) **Assertion :** Compound possessing same molecular formula but differing in their physical and chemical properties are called isomers.

Reason : Rearrangement reactions produce isomers.

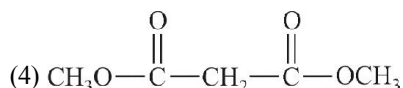
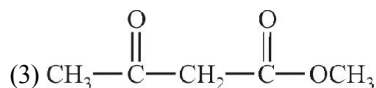
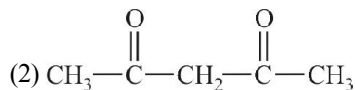
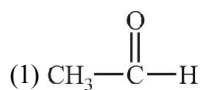
- (a) A (b) B
(c) C (d) D

2. (A) **Assertion :** Alkenes containing more than three carbons exhibit chain isomerism.

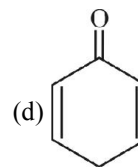
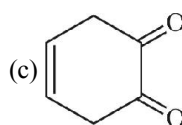
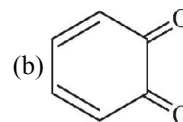
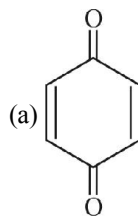
Reason : All the carbon atoms in alkenes are sp^3 –hybridized.

- (a) A (b) B
(c) C (d) D

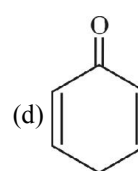
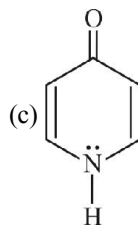
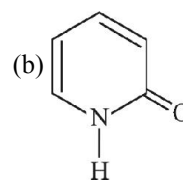
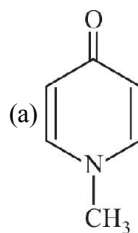
3. (I) Arrange in decreasing order of enolization of the following compounds



4. (M) Which can exhibit tautomerism here



5. (S) Which can not exhibit tautomerism here



6. (M) $\text{CH}_3\text{CH}_2\text{NO}_2$ will show isomerism

- (a) Position (b) Functional
(c) Tautomerism (d) Geometrical

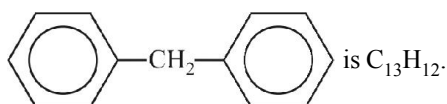
7. (M) HCN and HNC are..... isomers :

- (a) Functional (b) Tautomers
(c) Chain (d) Position

8. (S) How many possible carboxylic acids and esters are for a compound with molecular formula $C_4H_8O_2$?

- (a) four (b) six
(c) five (d) eight

9. (S) The molecular formula of diphenyl methane



How many structural isomers are possible when one of the hydrogen is replaced by a chlorine atom ?

- (a) 4 (b) 6
(c) 7 (d) 8

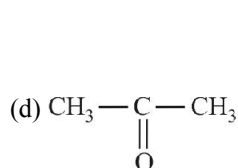
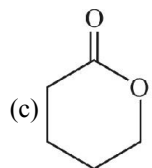
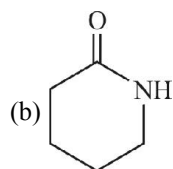
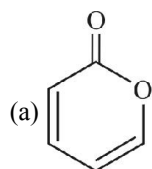
10. (S) Among the following alkenes

- (I) 1-Butene (II) cis-2-Butene
(III) trans-2-Butene

the decreasing order of stability is

- (a) III > I > II (b) I > II > III
(c) II > I > III (d) III > II > I

11. (S) Which one of the following compounds cannot show tautomerism ?



12. (A) **Assertion :** Structural isomers have different physical and chemical properties.

Reason : Structural isomers have different bonding pattern.

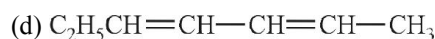
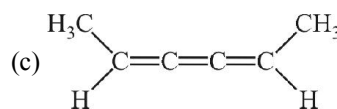
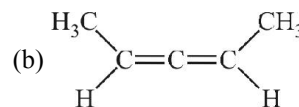
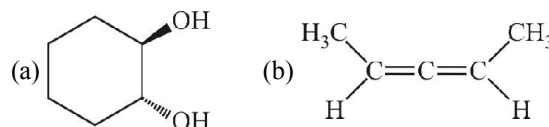
- (a) A (b) B
(c) C (d) D

13. (S) The degree of unsaturation or index of hydrogen deficiency in the following is:

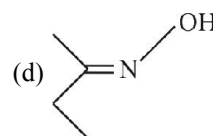
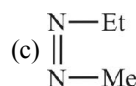
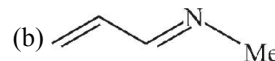
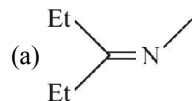
- (i) C_6H_{14} (ii) C_4H_6
(iii) C_6H_6
(a) 0, 2, 4 (b) 1, 0, 2
(c) 4, 1, 0 (d) 2, 0, 4

Geometrical Isomers

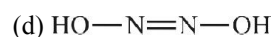
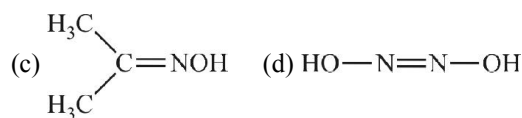
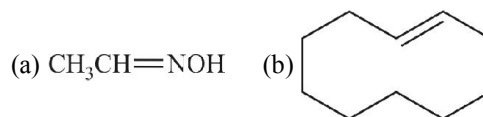
14. (M) Which of the following compound can exhibit geometrical isomerism ?



15. (M) Which of these compound can exhibit geometrical isomerism ?



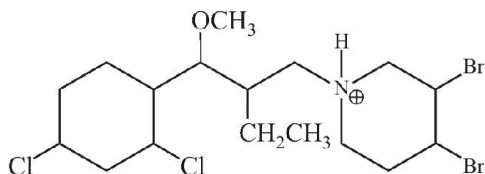
16. (M) Which of the following can show geometrical isomerism ?



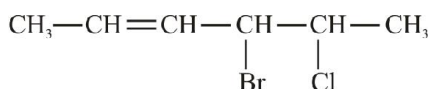
(c) $II > I > III > IV$ (d) $III > IV > II > I$

Stereocentres

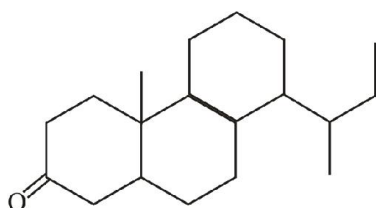
27. (I) How many chirality centers are present in the following molecule



28. (I) Find out the total number of stereocentre in the given compound.

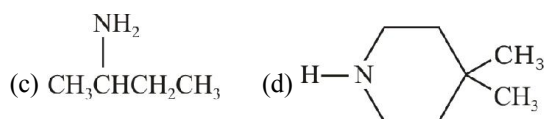
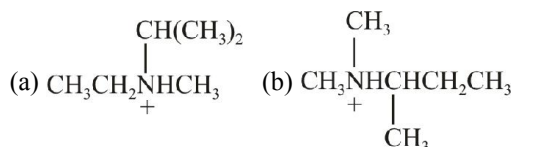


29. (S) Count the number of stereocentres in the molecule below



- (a) three (b) five
(c) six (d) seven

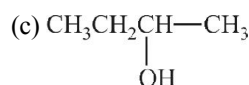
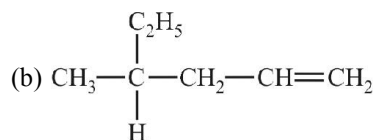
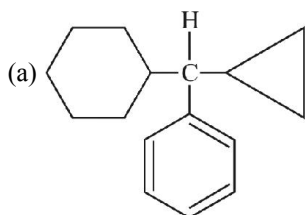
30. (S) Which of the following amines have a stereogenic centre?



- (a) I, II, III (b) III, IV, V
(c) I, III, IV (d) II, III, IV

Chiral Compounds

31. (S) Which of the following can not show optical activity ?



32. (M) An organic molecule necessarily shows optical activity if it

- (a) contains asymmetric carbon atoms
(b) is non-superimposable on its mirror image
(c) is superimposed on its mirror image.
(d) is chiral

33. (A) **Assertion** 1,3-dichloro propadiene rotate plane polarized light.

Reason The molecules of 1,3-dichloro propadiene are dissymmetric

- (a) A (b) B
(c) C (d) D

34. (S) Which of the following compounds is optically active?

- (a) 1-bromobutane (b) 2-bromobutane
(c) 1-bromo-2-methylpropane
(d) 2-bromo-2-methylpropane

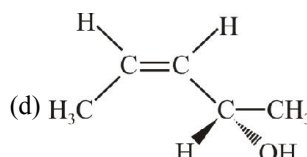
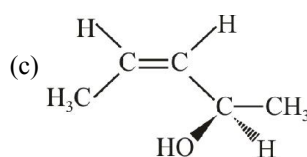
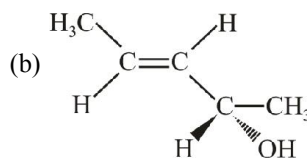
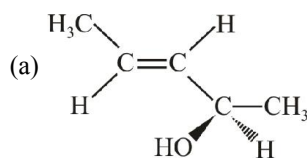
Optical Activity

35. (M) Which of the following statements is true ?

- (a) Doubling the concentration of the solution doubles the observed and specific rotation
(b) Doubling the concentration of the solution halves the observed and specific rotation
(c) Doubling the concentration of the solution doubles the observed rotation without affecting the specific rotation
(d) Doubling the concentration of the solution and doubling the length of the tube increases the observed rotation by four times.

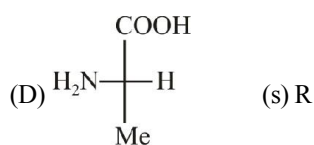
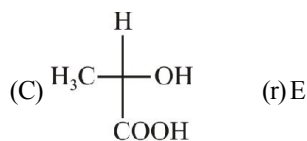
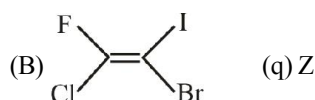
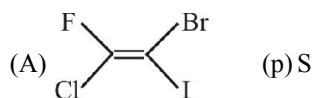
R/S Configuration

36. (S) Which of the following molecules have (2R, 3-Z) configuration?



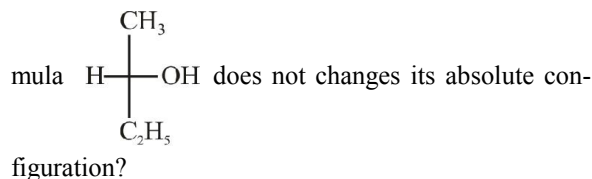
37. (X)

Column I	Column II
Compounds	Characteristics



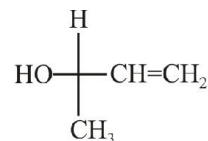
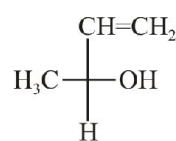
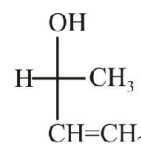
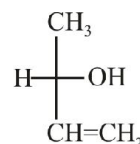
Fischer Projection

38. (S) Which of the following operations on the Fischer formula



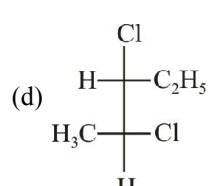
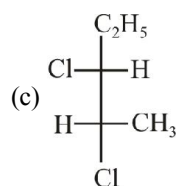
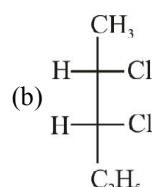
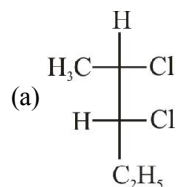
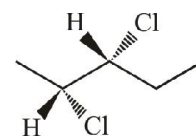
- Exchanging groups across the horizontal bond
- Exchanging groups across the vertical bond
- Exchanging groups across the horizontal bond and also across the vertical bond
- Exchanging a vertical and horizontal group

39. (S) Which of the following combinations amongst the four Fischer projections represents the same absolute configurations?

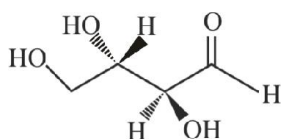


- II and III
- I and IV
- II and IV
- III and IV

40. (S) The Fischer projection that represents enantiomer of the given sawhorse projection is



41. (S) Which of the following switching or rotating Fischer structures changes its absolute configuration?
- An even number of switches
 - An odd number of switches
 - Rotating the Fischer projection by 180° in the plane of the paper
 - Exchange ligands across the horizontal as well as those across the vertical bond
42. (S) Which of the Fischer projection formula correspond to the following stereostructure?



- (a)
- (b)
- (c)
- (d)

Stereoisomers

43. (M) Select correct statements :

- (a) and both give colour with neutral FeCl_3 solution.
- (b) 2-pentanone and 3-pentanone are position as well as metamers
- (c) Product formed when benzaldehyde reacts with hydroxylamine shows geometric isomerism
- (d) 1, 2-dibromocyclohexane shows geometrical and optical isomerism

44. (A) **Assertion :** Meso-tartaric acid is optically active.

Reason : Optically active molecule is a molecule that cannot be superimposed on its mirror image

- (a) A (b) B
(c) C (d) D

45. (I) Calculate the total number of dichloro derivatives of cyclobutane. Fill the answer in the box.

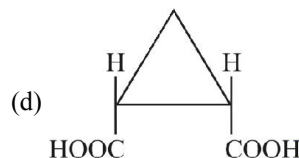
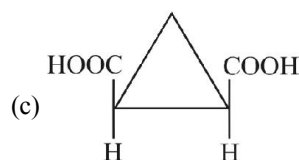
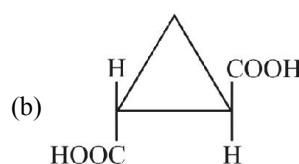
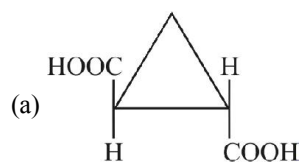
X	Y	Z
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X = Total number of dichloro derivatives.

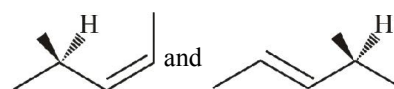
Y = Total number of optically active products.

Z = Total number of meso forms.

46. (M) Which of the following represent a pair enantiomers ?



47. (S) Following pair of molecules are related as



- (a) enantiomers (b) diastereomers
(c) identical (d) constitutional isomers

48. (A) **Assertion :** Enantiomers differ in their chemical action with other enantiomer.

Reason : A pair of enantiomers have different orientation of collision with another enantiomer forming different transition state.

- (a) A (b) B
(c) C (d) D

49. (A) **Assertion :** *Meso* form of 2, 3-butanediol has permanent optical inactivity.

Reason : It has internal enantiomerism

(a) A (b) B

(c) C (d) D

50. (A) **Assertion :** Both *cis*-1,3-Dimethyl cyclobutane and *trans*-1,3-dimethyl cyclobutane are optically inactive.

Reason : *cis*-1,3-dimethyl cyclobutane has the plane of symmetry, whereas *trans* form has the centre of symmetry.

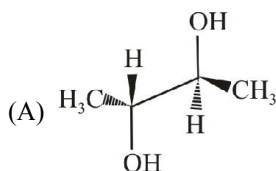
(a) A (b) B

(c) C (d) D

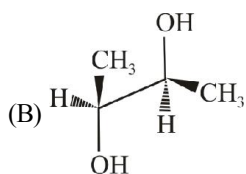
51. (X) Match the structures in column I with the properties in column II.

Column-I

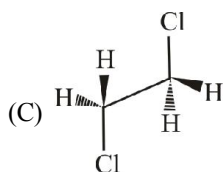
Column-II



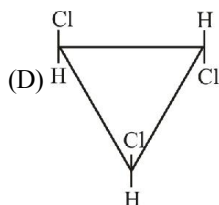
(p) Has plane of symmetry



(q) Has axis of symmetry



(r) Has centre of symmetry



(s) Rotate plane polarized light

52. (X) Match the molecule in column A with type of stereoisomerism shown by them in column B.

Column-I

Column-II

(A) 2,3-dichlorobutane

(p) Enantiomerism

(B) 2-methyl-3-pentenoic acid

(q) Diastereomerism

(C) 2-butanol

(r) Meso form

(D) 1,3-dichloro propadiene

(s) Conformational

53. (X) Match the terms in column A with properties in column B.

Column-A

Column-B

(A) Enantiomers

(p) Can be separated into enantiomers

(B) Diastereomers

(q) Have different actions on enantiomer

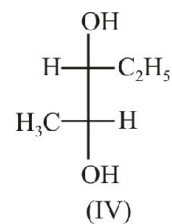
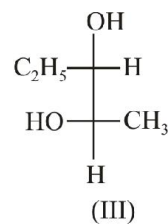
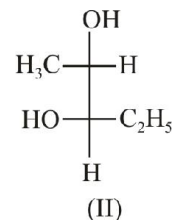
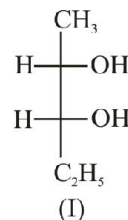
(C) Racemic mixture

(r) Have different action on achiral compounds

(D) Meso form

(s) Have permanent optical inactivity

54. Consider the following Fischer projections and match the pair in column A with the relation in column B.



Column-I

Column-II

(A) I + II

(p) Enantiomers

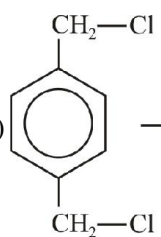
(B) I + III

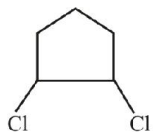
(q) Diastereomers

(C) II + IV

(r) Identical

(D) I + IV

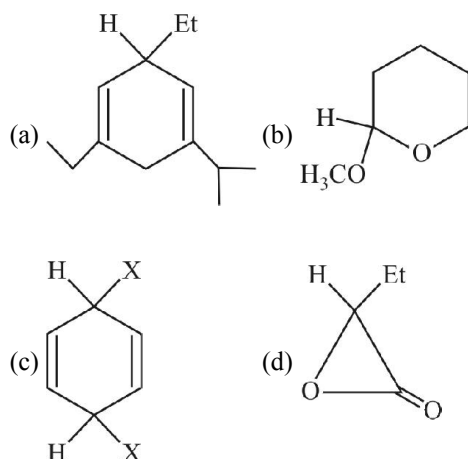
55. (I) (a)  $\longrightarrow$ (x) (Number of plane of symmetry)

(b)  $\longrightarrow$ (y) (Number of mesoisomer)

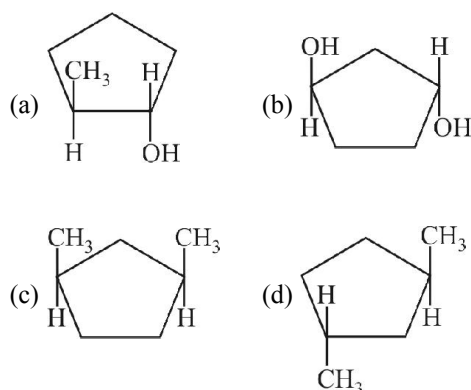
Sum of $(x + y) = ?$

Chirality in Cyclic Compounds

56. (M) Identify the compound, which has a stereocenter here.



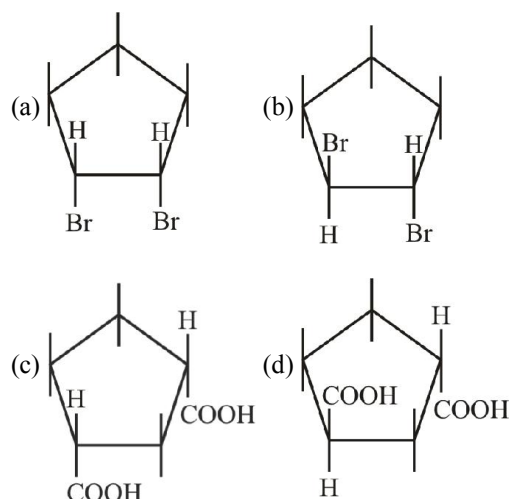
57. (M) Which of the following cyclopentane derivative is/are optically active ?



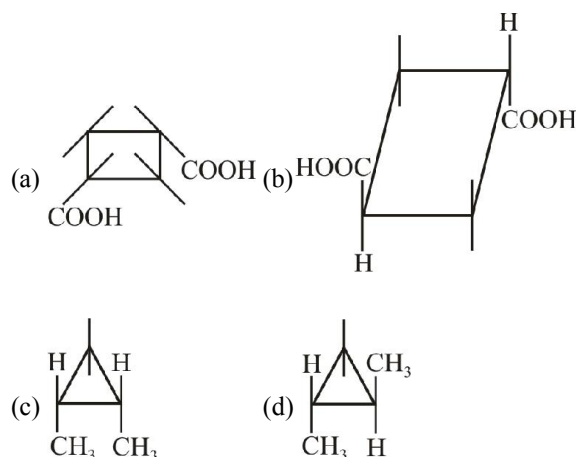
58. (M) Which dimethylcyclobutane is/are optically inactive ?

- (a) trans-1, 2 (b) trans-1, 3
(c) cis-1, 2 (d) cis-1, 3

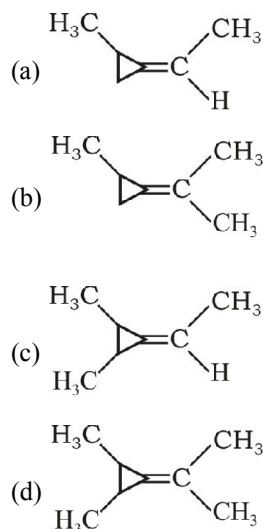
59. (M) Which out the following are non-resolvable?



60. (S) Which of the following are resolvable?



61. (M) Which of the following will show optical isomerism as well as geometrical isomerism?



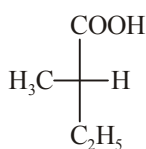
Racemization

62. (S) Which of the following on standing in dilute, aqueous basic solution will undergoes racemization?

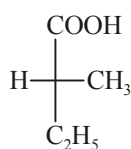
- (a) $\text{Ph-CHBr-CH}_2\text{-CHO}$
- (b) PhCHBr-CHCl-CHO
- (c) $\text{Ph-CH}_2\text{-CHBr-CHO}$
- (d) Ph-CHClCHBr-CHO

Comprehension

A racemic mixture containing following enantiomers are to be separated into pure enantiomers by reacting with a pure enantiomer.

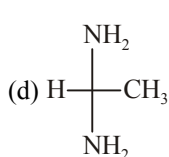
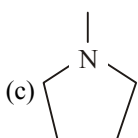
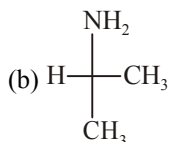
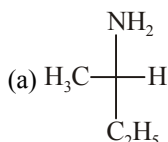


I



II

63. (C) Which of the following amines can be used for the separation of the above-mentioned racemic mixture?



64. (C) What would be the stereometric relations between the products formed when the above-mentioned racemic mixture is allowed to react with desired amine of the previous question in order for separation of racemic mixture?

- (a) The products would be identical
- (b) The products would be a pair of enantiomer
- (c) The products would be a pair of diastereomers
- (d) The products would not be stereoisomers

Comprehension

A solution contain three stereoisomers of 2,3-dihydroxy butanedioic acid. On fractional crystallization of this solution, two fractions. **Fraction-I** and **Fraction-II** are separated. Fraction-I on reaction with (+)-2-butanamine gives a single salt while Fraction-II on reaction with the same (+)-2-butanamine gives a mixture of two salts. Answer the following three questions based on the above information.

65. (C) What can be accurately predicted about the molecules of fraction-I?

- (a) Molecules of fraction-I are enantiomers
- (b) Molecules of fraction-I are meso compound
- (c) Molecules of fraction-I are in racemic form
- (d) Molecules of fraction-I has no chiral centre

66. (C) What is the relationship between the molecules of fraction-I with the molecules of fraction-II

- (a) They are enantiomers
- (b) They are identical
- (c) They are diastereomers
- (d) They are diastereomers as well as optically active

67. (C) What is the relationship between the two different salts obtained from fraction-II?

- (a) They are structural isomers
- (b) They are enantiomers
- (c) They are diastereomers
- (d) They are conformers

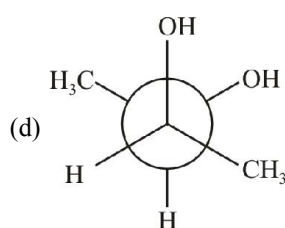
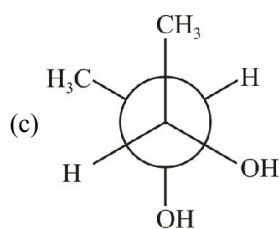
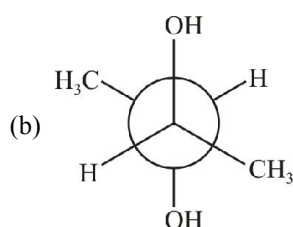
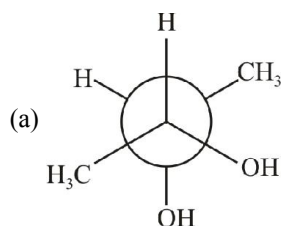
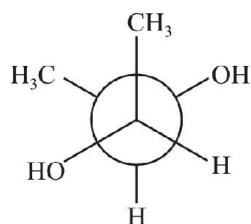
Conformers

68. (A) **Assertion** : Cyclobutane is less stable than cyclopentane.

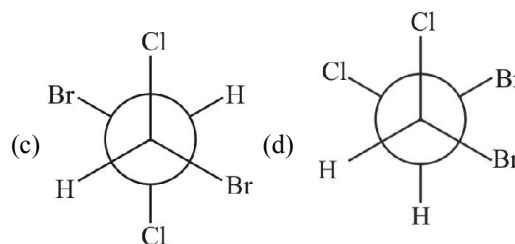
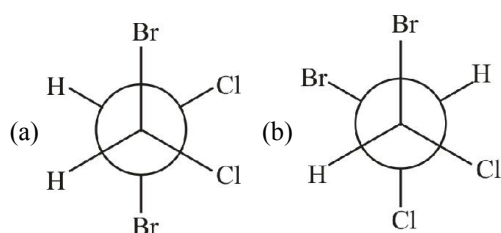
Reason : Presence of 'bent bond' causes loss of orbital overlap.

- (a) A
- (b) B
- (c) C
- (d) D

69. (M) Which of the following molecules is identical with that represented by



70. (S) The most stable conformer of meso-1,2-dibromo-1,2-dichloro ethane is

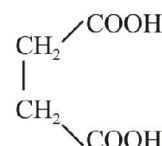
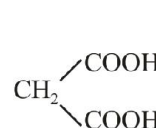
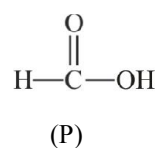


Total Number of Isomers

Comprehension

An organic compound having only C, H atoms has a molecular wt. 86. By nature it is a liquid compound and it is capable of showing chain, position isomerism. It can also show free radical halogenation reaction also.

71. (C) How many structural isomers of C_6H_{14} are possible ?
(a) 3 (b) 4
(c) 5 (d) 6
72. (C) How many of its structural isomers is/are optically active ?
(a) Zero (b) 1
(c) 2 (d) 3
73. (C) Which of the following alkane on single mono chlorination can give 3-mono substituted products ?
(a) 2-Methylpentane (b) n-Hexane
(c) 2,2-Dimethylbutane
(d) both (b) & (c)
74. (S) How many open chain structural isomers of compound with molecular formula C_6H_{12} show geometrical isomerism ?
(a) 2 (b) 3
(c) 4 (d) 5
75. (S) Write the correct order of acidity:



- (a) $P > Q > R > S$ (b) $Q > P > R > S$
(c) $Q > R > S > P$ (d) $S > R > Q > P$

76. (S) Number of isomers which can be obtained theoretically on monochlorination of 2-methylbutane is

- (a) 2 (b) 3
(c) 4 (d) 5

77. (S) How many structural (including stereoisomers) isomers could be obtained by replacing one hydrogen of propene with chlorine ?

- (a) 2 (b) 3
(c) 4 (d) 5

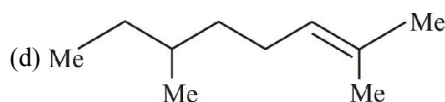
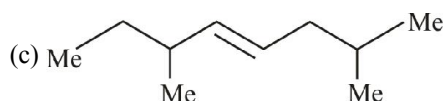
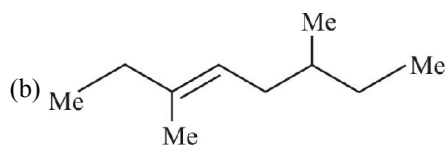
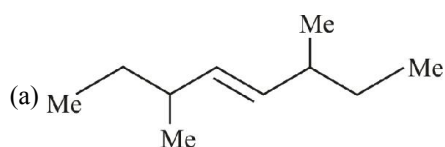
78. (S) Number of stereoisomers of the compound 2-chloro-4-methylhex-2-ene is

- (a) 2 (b) 4
(c) 8 (d) 16

Comprehension

An organic compound (A) ($C_{10}H_{20}$) on reductive ozonolysis gives 2-methyl butanal. Based on this information, answer the following questions.

79. (C) Compound (A) is



80. (C) The total number of stereoisomers possible for (A) is:

- (a) 8 (b) 7
(c) 6 (d) 4

81. (C) If the stereochemistry about the double bond in (A) is cis, the total number of stereoisomers for (A) is:

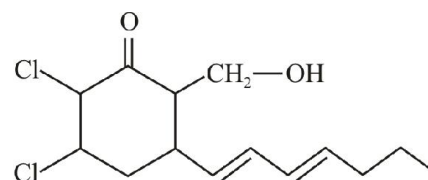
- (a) 2 (b) 3
(c) 4 (d) 5

82. (C) If the stereochemistry about the double bond in (A) is cis, the total number of enantiomers for (A) is

- (a) None (b) 2
(c) 4 (d) 6

83. (I) How many different isomers (including stereo isomers) of a monocarboxylic acid with molecular formula $C_6H_{12}O_2$, on decarboxylation with soda lime will produce the same n-pentane?

84. (I)

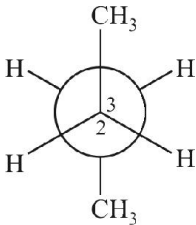


Number of stereoisomer are

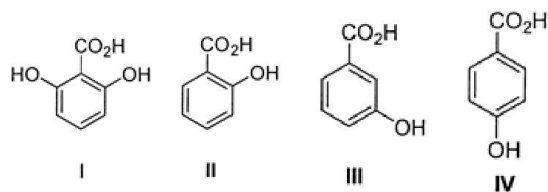
85. (I) Total number of isomers for $C_4H_6Br_2$ containing cyclobutane ring are (including stereoisomer) ?

EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

Objective Questions (Only one correct option)

1. The compound which is not isomeric with diethyl ether is (1981)
 (a) n-Propyl Methyl Ether (b) Butan-1-ol
 (c) 2-Methyl Propan-2-ol (d) Butanone
2. Which of the following compounds will exhibit cis-trans (geometrical) isomerism ? (1983)
 (a) But-2-ene (b) But-2-yne
 (c) Butan-2-ol (d) Butanal
3. An isomer of ethanol is (1986)
 (a) methanol (b) diethyl ether
 (c) acetone (d) dimethyl ether
4. Which of the following will have least hindered rotation about carbon-carbon bond ? (1987)
 (a) Ethane (b) Ethylene
 (c) Acetylene (d) Hexachloroethane
5. The number of isomers of C_6H_{14} is (1987, 2007)
 (a) 4 (b) 5
 (c) 6 (d) 7
6. The enolic form of acetone contains (1990)
 (a) 9 sigma bonds, 1 pi bond and 2 lone pairs
 (b) 8 sigma bonds, 2 pi bonds and 2 lone pairs
 (c) 10 sigma bonds, 1 pi bond and 1 lone pair
 (d) 9 sigma bonds, 2 pi bonds and 1 lone pair
7. Isomers which can be interconverted through rotation around a single bond are (1992)
 (a) conformers (b) diastereomers
 (c) enantiomers (d) positional isomers
8. How many optically active stereoisomers are possible for butane-2, 3-diol ? (1997)
 (a) 1 (b) 2
 (c) 3 (d) 4
9. The optically active tartaric acid is named as D-(+)-tartaric acid because it has a positive (1999)
 (a) optical rotation and is derived from D-glucose
 (b) pH in organic solvent
 (c) optical rotation and is derived from D-(+)-glyceraldehyde
 (d) optical rotation when substituted by deuterium
10. Which of the following compounds will exhibit geometrical isomerism ? (2000)
 (a) 1-phenyl-2-butene (b) 3-phenyl-1-butene
 (c) 2-phenyl-1-butene (d) 1, 1-diphenyl-1-propene
11. The number of isomers for the compound with molecular formula $C_2BrClFI$ is (2001)
 (a) 3 (b) 4
 (c) 5 (d) 6
12. Which of the following compounds exhibits stereoisomerism ? (2002)
 (a) 2-Methyl But-1-ene
 (b) 3-Methyl But-1-yne
 (c) 3-Methyl Butanoic Acid
 (d) 2-Methyl Butanoic Acid
13.  (2004)
 C_2 is rotated clockwise 120° about C_2-C_3 bond. The resulting conformer is
 (a) partially eclipsed (b) eclipsed
 (c) gauche (d) staggered
14. The number of stereoisomers obtained by bromination of trans-2-butene is (2007)
 (a) 1 (b) 2
 (c) 3 (d) 4

15. The correct order of acidity for the following compounds is (2013)



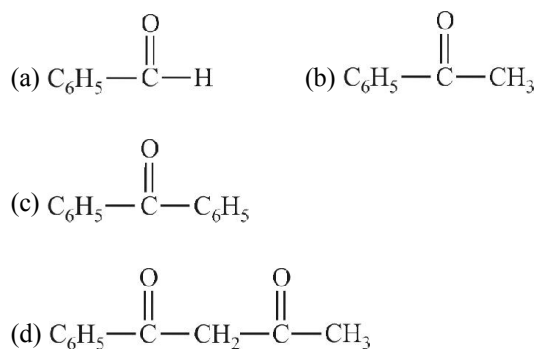
- (a) I > II > III > IV (b) III > I > II > IV
(c) III > IV > II > I (d) I > III > IV > II
16. **Statement-1** : Molecules that are non-superimposable on their mirror images are chiral. (2007)

Statement-2 : All chiral molecules have chiral centres.

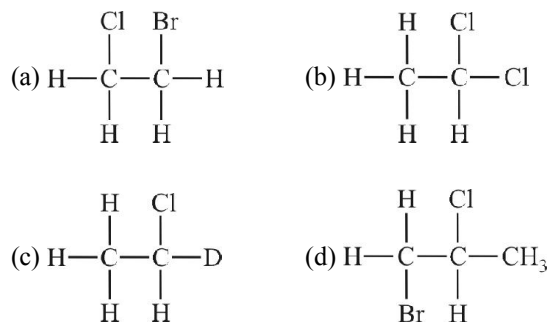
- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Objective Questions (One or more than one correct option)

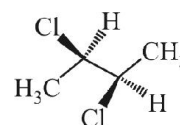
17. Only two isomeric monochloro derivatives are possible for (1986)
- (a) n-Butane (b) 2, 4-Dimethylpentane
(c) Benzene (d) 2-Methylpropane
18. Keto-enol tautomerism is observed in (1988)



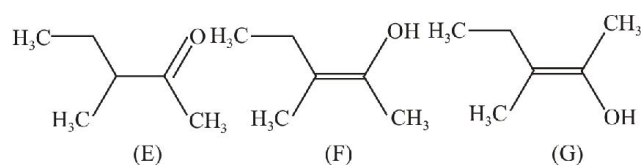
19. Which of the following have asymmetric carbon atom? (1989)



20. The correct statement(s) about the compound given below is (are) (2008)

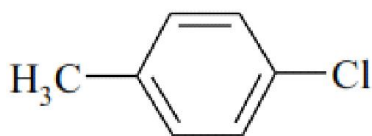


- (a) the compound is optically active
(b) the compound possesses centre of symmetry
(c) the compound possesses plane of symmetry
(d) the compound possesses axis of symmetry
21. The correct statement(s) concerning the structures E, F and G is (are) (2009)



- (a) E, F and G are resonance structures
(b) E, F and E, G are tautomers
(c) F and G are geometrical isomers
(d) F and G are diastereomers
22. The correct statement (s) about the compound $\text{H}_3\text{C}(\text{HO})\text{HC}=\text{CH}=\text{CH}(\text{OH})\text{CH}_3$ (X) is (are) (2009)
- (a) The total number of stereoisomers possible for X is 6.
(b) The total number of diastereomers possible for X is 3.
(c) If the stereochemistry about the double bond in X is trans, the number of enantiomers possible for X is 4.
(d) If the stereochemistry about the double bond in X is cis, the number of enantiomers possible for X is 2.

23. The IUPAC name(s) of the following compound is (are) (2017)



- (a) 4-chlorotoluene
(b) 4-methylchlorobenzene
(c) 1-chloro-4-methylbenzene
(d) 1-methyl-4-chlorobenzene

Fill in the Blanks

24. The IUPAC name of succinic acid is (1994)
25. ring is most strained. (Cyclopropane, Cyclobutane, Cyclopentane). (1981)
26. Isomers which are mirror images are known as (superimposable, non-superimposable, enantiomers, diastereomers, epimers) (1988)

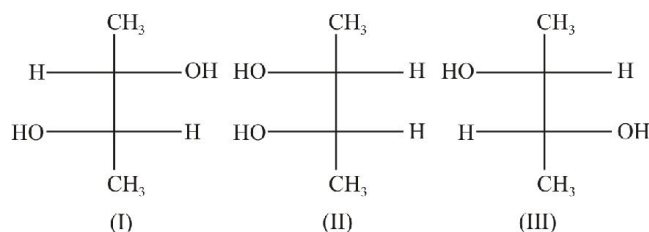
True/False

27. m-chlorobromobenzene is an isomer of m-bromochlorobenzene. (1985)
28. 2, 3, 4-trichloropentane has three asymmetric carbon atoms. (1990)

Subjective Questions

29. Write structural formulae for the isomeric alcohols having the molecular formula $C_4H_{10}O$. (1984)
30. Write the structure of all the possible isomers of dichloroethene. Which of them will have zero dipole moment? (1985)

31. Write tautomeric forms of phenol. (1992)
32. Identify the pairs of enantiomers and diastereomers from the following (2000)



33. (a) Draw Newman's projection for the less stable staggered form of butane.
(b) Relatively less stability of the staggered form is due to
(i) Torsional strain
(ii) van der Waals' strain
(iii) Combination of the above two

$$34. \mu_{\text{obs}} = \sum \mu_i x_i$$

where μ_i is the dipole moment of stable conformer and x_i is the mole fraction of that conformer.

- (a) Write stable conformer for $Z - CH_2 - CH_2 - Z$ in Newman's projection.

If $\mu_{\text{solution}} = 1.0\text{D}$ and mole fraction of anti form = 0.82, find μ_{gauche}

- (b) Write most stable meso conformer of $(CHDY)_2$

If (i) $Y = CH_3$ about $C_2 - C_3$ rotation and

(ii) $Y = OH$ about $C_1 - C_2$ rotation. (2005)

35. Give the total number of cyclic structural as well as stereo isomers possible for a compound with the molecular formula C_5H_{10} . (2009)

ANSWER KEY**EXERCISE - 1 : (Basic Objective Questions)**

1. (a)	2. (b)	3. (b)	4. (d)	5. (c)	6. (b)	7. (b)	8. (a)	9. (c)
10. (b)	11. (c)	12. (b)	13. (c)	14. (c)	15. (d)	16. (c)	17. (d)	18. (b)
19. (c)	20. (c)	21. (d)	22. (d)	23. (d)	24. (a)	25. (b)	26. (d)	27. (d)
28. (a)	29. (c)	30. (c)	31. (d)	32. (b)	33. (a)	34. (d)	35. (b)	36. (d)
37. (c)	38. (c)	39. (c)	40. (b)	41. (d)	42. (b)	43. (c)	44. (a)	45. (c)
46. (a)	47. (c)	48. (c)	49. (c)	50. (c)	51. (d)	52. (b)	53. (c)	54. (d)
55. (d)	56. (d)	57. (b)	58. (c)	59. (b)	60. (c)	61. (d)	62. (c)	63. (b)
64. (b)	65. (d)	66. (d)	67. (c)	68. (b)	69. (a)	70. (a)	71. (b)	72. (c)
73. (c)	74. (a)	75. (c)	76. (b)	77. (d)	78. (b)	79. (c)	80. (b)	81. (a)
82. (b)	83. (c)	84. (a)	85. (b)					

EXERCISE - 2 : (Previous Year JEE Mains Questions)

1. (d)	2. (d)	3. (b)	4. (c)	5. (c)	6. (c)	7. (c)	8. (c)	9. (b)
10. (a)	11. (a)	12. (c)	13. (c)	14. (a)	15. (c)	16. (b)	17. (d)	18. (b)
19. (d)	20. (d)	21. (a)						

JEE Mains Online

1. (a)	2. (d)	3. (c)	4. (b)	5. (d)	6. (c)	7. (c)	8. (b)	9. (a)	10. (a)
11. (b)	12. (c)	13. (b)	14. (a)	15. (c)	16. (c)\	17. (b)	18. (d)	19. (c)	

EXERCISE - 3 : (Advanced Objective Questions)

- | | | | | | | | | |
|--|------------|------------|------------|------------------------------|-----------|------------------------------------|------------------------|------------|
| 1. (b) | 2. (c) | 3. (2341) | 4. (cd) | 5. (a) | 6. (bc) | 7. (ab) | 8. (b) | 9. (a) |
| 10. (d) | 11. (a) | 12. (a) | 13. (a) | 14. (acd) | 15. (bcd) | 16. (abd) | 17. (cd) | 18. (b) |
| 19. (a) | 20. (b) | 21. (b) | 22. (abd) | 23. (ab) | 24. (b) | 25. (c) | 26. (b) | 27. (0008) |
| 28. (0003) | 29. (c) | 30. (a) | 31. (d) | 32. (bd) | 33. (a) | 34. (b) | 35. (cd) | 36. (c) |
| 37. A-q, B-r, C-s, D-p | 38. (c) | 39. (c) | 40. (a) | 41. (b) | 42. (b) | 43. (abcd) | 44. (d) | |
| 45. (0621) | 46. (ab) | 47. (b) | 48. (a) | 49. (a) | 50. (a) | 51. A-p, r; B-q, s; C-p, q, r; D-p | | |
| 52. A-p, q, r, s; B-p, q, s; C-p, s; D-p | | | | 53. A-q; B-q, r; C-p, q; D-s | | | 54. A-r; B-p, C-q, D-q | |
| 55. (0004) | 56. (abd) | 57. (abd) | 58. (bcd) | 59. (ac) | 60. (d) | 61. (acd) | 62. (c) | 63. (a) |
| 64. (c) | 65. (b) | 66. (c) | 67. (c) | 68. (a) | 69. (ad) | 70. (c) | 71. (c) | 72. (a) |
| 73. (d) | 74. (c) | 75. (c) | 76. (d) | 77. (c) | 78. (b) | 79. (a) | 80. (c) | 81. (b) |
| 82. (b) | 83. (0004) | 84. (0064) | 85. (0006) | | | | | |

EXERCISE - 4 : (Previous Year JEE Advanced Questions)

- | | | | | | | | | |
|------------------|----------|-------------------------------------|------------|------------|-------------------------------------|----------|------------|------------|
| 1. (d) | 2. (a) | 3. (d) | 4. (a) | 5. (b) | 6. (a) | 7. (a) | 8. (b) | 9. (c) |
| 10. (a) | 11. (d) | 12. (d) | 13. (c) | 14. (a) | 15. (a) | 16. (c) | 17. (a, d) | 18. (b, d) |
| 19. (c, d) | 20. (ad) | 21. (b, c, d) | 22. (a, d) | 23. (a, c) | 24. Non-superimposable, Enantiomers | | | |
| 25. cyclopropane | | 26. non-superimposable, enantiomers | | | 27. False | 28. True | | |